

**DIGITAL SKILLS****Student Learning Outcomes:**

After the completion of this unit, students will be able to:

- Define operating system and identify its functions, discuss type of operating systems for computers and mobile phones, demonstrate the startup procedure of Windows operating system.
- Identify the desktop icons/tools and explain their functions, create, name, rename, delete, cut, copy and paste files and folders, drag and drop files/folders, and create a shortcut.
- Create, save, edit images using MS-Paint image processing software.
- Draw and colour freehand drawings, 2D shapes like squares, rectangles, circles, lines, etc., add text to a drawing, colour, outline and position drawings as needed.
- Define world wide web and web browser, list popular web browsers and activities performed using web browsers, describe search engines and demonstrate searching skills to look for certain topics.

**Introduction to the Unit:**

This unit covers different components of Windows 10 operating system and its functionality. It explores desktop and desktop icons, taskbar, start

button. It also explores how to manage the files and folders in Windows 10. It also explores image processing using MS-Paint. It also covers how to use the Internet to conduct a search query.

2.1 OPERATING SYSTEM AND ITS FUNCTIONS**2.1.1 Definition of Operating System:**

An operating system is a program that manages and controls all the activities of the computer system. It manages all the parts of the computer and its programs, making it simple and easy for us to use the computer.

2.1.2 Functions of an Operating System:

The most common functions of an operating system are shown in figure 2.1.



Figure 2.1 Functions of an operating system

An operating system manages computer resources such as memory, CPU, storage, and devices like the keyboard and mouse. It provides an easy way for you to interact with the computer using icons, buttons and menus. It helps run programs smoothly, organizes files and folders, and keeps your computer safe from viruses and unauthorized access.

2.2 TYPES OF OPERATING SYSTEMS

2.2.1 Operating Systems for Computers:

Some of the famous operating systems for computers include Windows, Mac OS, Linux and UNIX. Windows is an operating system developed by Microsoft Corporation. It is one of the most popular operating systems for personal computers. Windows

provides a graphical user interface (GUI) that allows users to interact with their computers easily.



Mac OS is an operating system developed by Apple Inc. It is designed exclusively for use on Apple Macintosh computers. Mac OS is known for its user-friendly interface, stability, and seamless integration with other Apple devices and services.

Linux is an open-source operating system derived from UNIX. It has many versions, called distributions. Linux is popular because of its flexibility, security, and user-friendly interface.

UNIX is an open-source operating system developed at AT&T Bell Labs in the USA. Linux and its various distributions are either derived from UNIX or influenced by it.

2.2.2 Operating Systems for Mobile Phones:

Most commonly used operating systems for mobile phones include Android, iOS and Windows 10 Mobile.

Android is developed by Google. It is the most widely used operating system for mobile phones. It supports the apps available through

the Google Play Store. iOS, developed by Apple, is exclusive to iPhones, iPads, and iPods. It is known for its user-friendly interface and easy integration with other Apple devices. It supports the apps available through Apple



App Store. Windows 10 Mobile is developed by Microsoft. It is designed for smartphones and tablets and supports Windows apps.

2.3 WINDOWS 10 STARTUP PROCEDURE

The windows startup procedure is a process of making computer ready for use. The steps of Windows 10 startup procedure are given in figure 2.2.

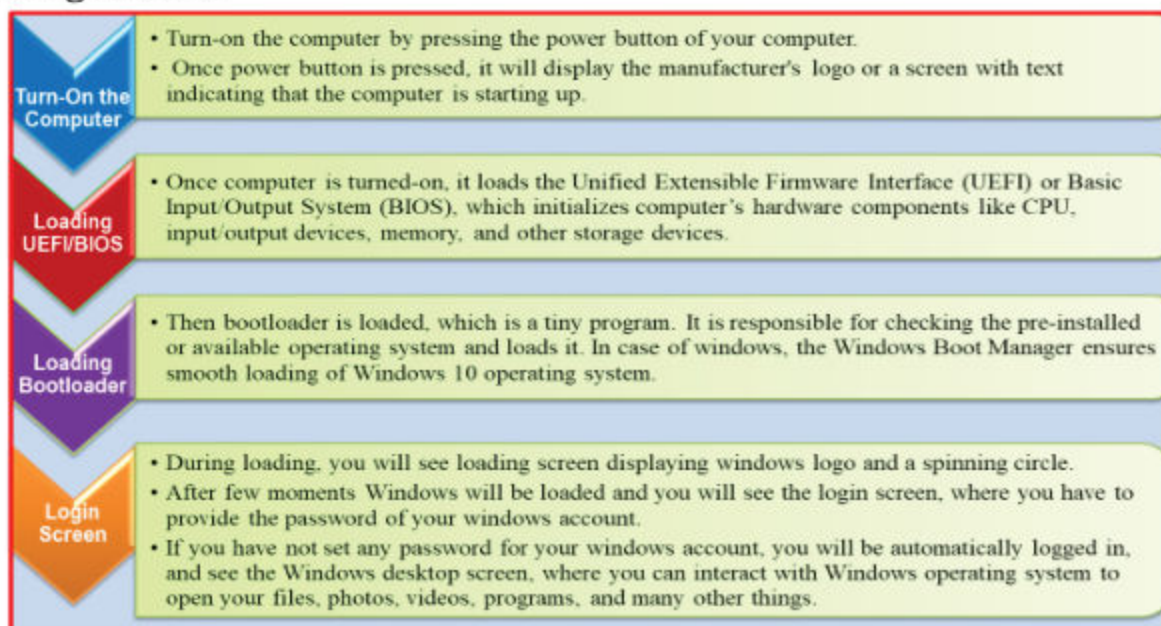


Figure 2.2 Windows 10 Startup Procedure

2.4 COMMON COMPONENTS OF WINDOWS 10

Figure 2.3 labels the main icons and tools on the Windows 10 Desktop, which are explained one by one in this section.

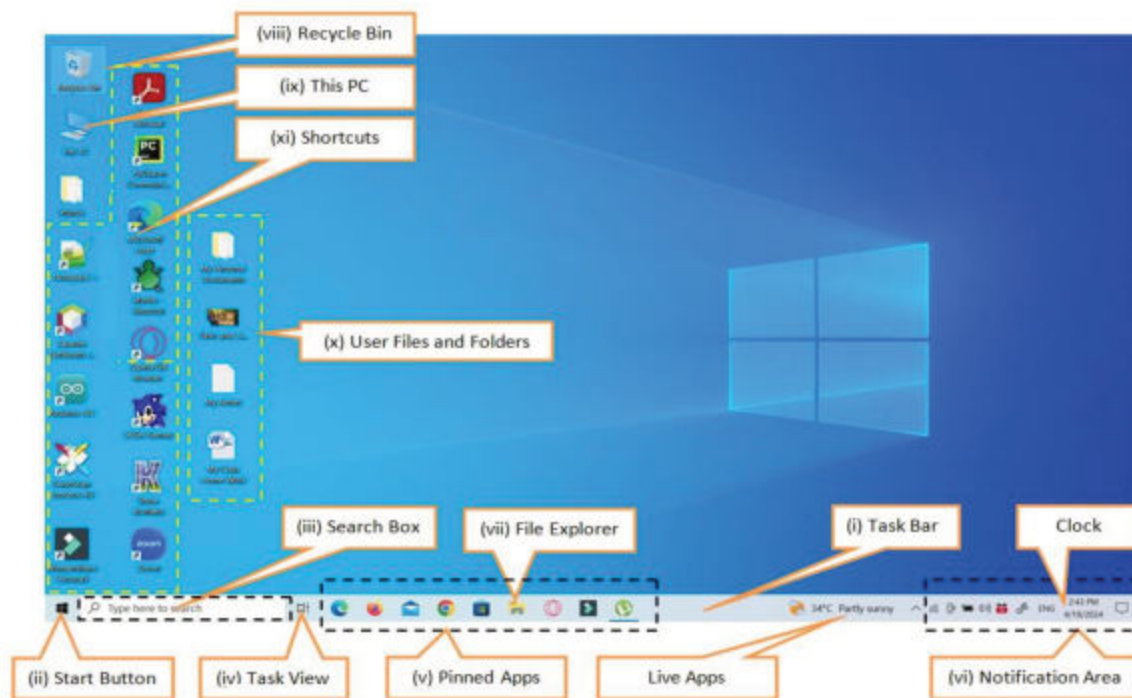


Figure 2.3 Windows 10 Desktop Components

(i) Task Bar:



The taskbar is available at the bottom of Windows 10 desktop. It is an organization tool that organizes and provides quick access to various Windows features ranging from quick search or opening of applications to viewing and setting date and time, etc.

(ii) Start Button:

The Windows start button is also known as the Windows icon or Windows button. It is the main navigation tool that is available on bottom left corner on the Windows 10 taskbar. By clicking on this button, you can open the Start menu that provides quick access to various Windows 10 features, like power options, system settings, accessing files, accessing user account options, application and system programs, recently used apps and documents, etc.



❖ ACTIVITY

Exploring the Start Button and Power Options

Watch and Learn: Your teacher will show you how to use the Start button, especially the Power Options like Shut Down, Restart, and Sleep.

Hands-On Practice: After the demonstration, you will get a chance to try it yourself. Your task will be to either power off, restart, or put your computer into sleep mode.

(iii) Search Box:

The search box is next to the start button on the taskbar.



In search box you can type

keywords to search for photos, videos, files, folders, applications, and settings. You can also search items on web using this search box. The good thing of this search box is that it will provide you instant results as you start typing something in it that makes it easy for you to find what you need quickly.

(iv) Task View:

Next to the search box, there is a tiny button called Task View. Task view button allows you to switch between open windows and virtual desktops. It gives you a quick way to manage and organize multiple tasks as well as multiple workspaces.



(v) Pinned Apps/Quick Launch:



Pinned apps are also known as “taskbar shortcuts” or “quick launch”. Pinned apps on the taskbar provide quick and easy way to access the frequently used programs. By pinning an application (like a web

browser) to the taskbar, you can easily launch it with a single click instead of searching for it using the search box or in start menu.

ACTIVITY

Customizing Your Taskbar with Pinned Apps

Watch and Learn: Your teacher will show you how to add and remove apps in the quick launch/pinned apps area of the taskbar.

Hands-On Practice: After the demonstration, you will get a chance to customize your own taskbar by pinning your favourite applications like games, typing software etc

(vi) System Tray/Notification Area:

System tray also known as notification area is located at the bottom right corner of the taskbar. It shows icons for



applications and system notifications. It also displays system status indicators like battery level, volume control, network connectivity, and date/time. You can click on these icons to access settings and perform some actions on it like modifying date and time.

(vii) File Explorer:

File explorer icon is located on the taskbar in pinned apps/quick launch area. You can browse and manage files and folders on your computer using the File Explorer (as

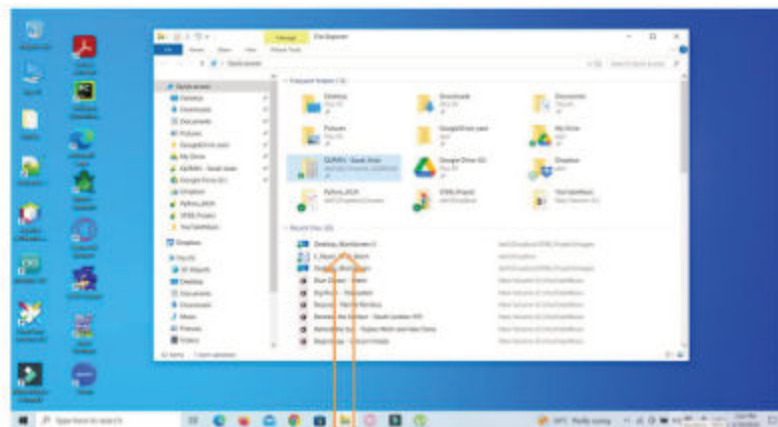


Figure 2.4 Windows 10 File Explorer

shown in Figure 2.4). It allows you to create, copy, move, rename, and delete files and folders. It also allows you access to various drives and storage locations, such as CD Drives, USB Drives and other Storage Devices.

(viii) Recycle Bin

Recycle Bin is a special type of system folder in Windows that temporarily stores all deleted files and folders. You can restore or recover deleted items from the recycle bin. However, any file or folder deleted from the recycle bin cannot be recovered.



(ix) This PC (My Computer)

The PC / My Computer icon (as shown in figure 2.5) provide you quick access to all the drives and storage devices connected to your computer such as local disk drives (e.g. C, D, E, etc), external hard drives, cloud storages (e.g. Google drive, Dropbox, etc), USB drives, and network drives.



Note

If you have deleted any file or folder using CTRL+SHIFT+DELETE shortcut key, it cannot be recovered.

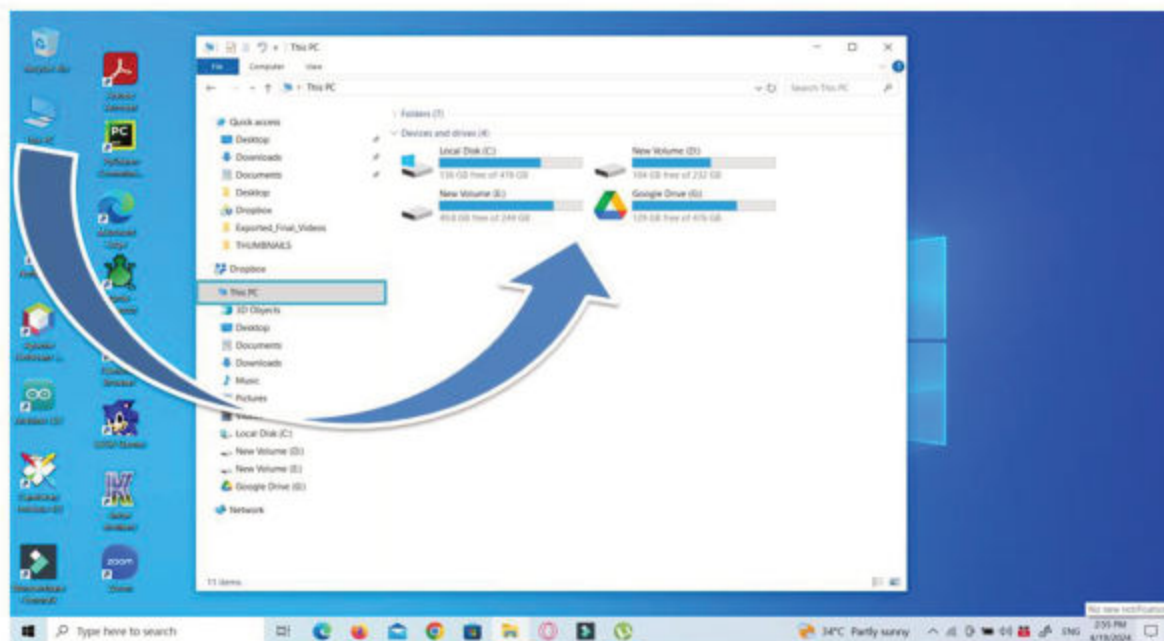
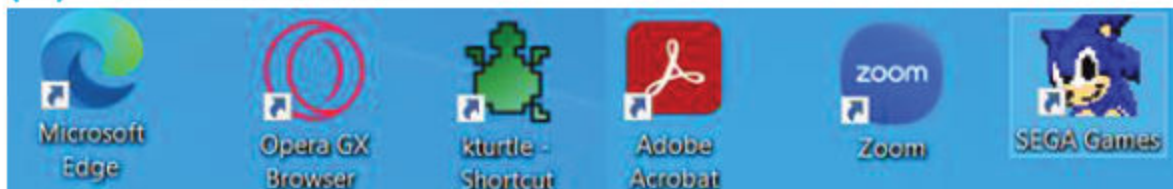


Figure 2.5 This PC (My Computer) Content

(x) User Files

Sometime users also store different types of files on the desktop. Common user files include downloaded web pages, word documents, excel spread sheets, pdf files, photos, audio and video files, and text files. These files are identified by their icons. Windows assigns each category of files a different icon as shown below.

**(xi) Shortcut Icons:**

A shortcut icon represents a specific program or application that is installed on your computer. An upward arrow sign indicates that it is a shortcut icon. By double clicking the shortcut icon, you can launch its corresponding application or program. By using shortcuts, you can access your favourite programs and applications in a quick and easy manner.

ACTIVITY

Exploring the Desktop

Watch and Learn: Your teacher will show you the desktop and explain the different items available on it.

Hands-On Practice: After the demonstration, you will explore the functions of various desktop items like the start menu, search box, quick launch, and notification area.

2.5 WORKING WITH WINDOWS 10

2.5.1 How to open a file or run an application program?

In Windows 10, you can open a file or run an application program from the desktop, the start menu, or the search box. Each of these methods is explained below.

(i) Open a file or run an application program from desktop

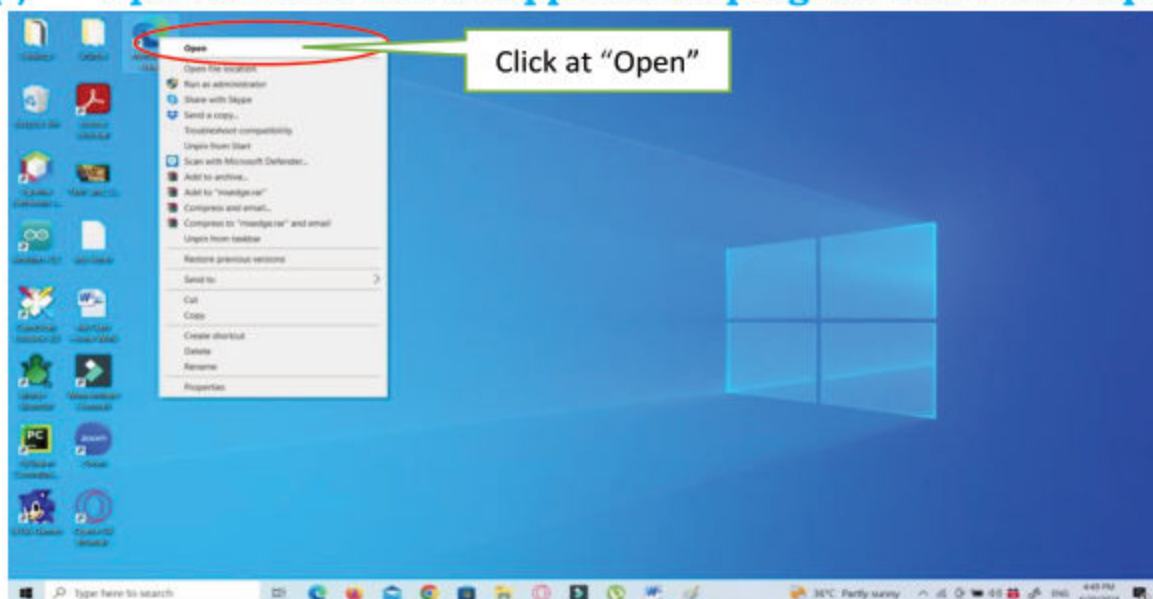


Figure 2.6 Open a file or run an application from desktop

You can see different types of icons on Windows 10 desktop. Each of these icons represents either a file/folder or an application program. You can open any of these files/folders or run any of these applications just by double clicking on the icon. Alternatively, you can also open a file/folder or run an application by right clicking on

its associated icon, and then clicking on “open” option in popup menu as shown in figure 2.6.

(ii) Open a file or run an application program from start menu

You can also open a file or run an application program from the start menu. The start menu contains the list of programs installed on the computer as well as recently opened files. To open any file or run any program from the start menu, simply press start/window button at the left bottom corner of the desktop/taskbar, and then click on the desired file or program name as illustrated in Figure 2.7.

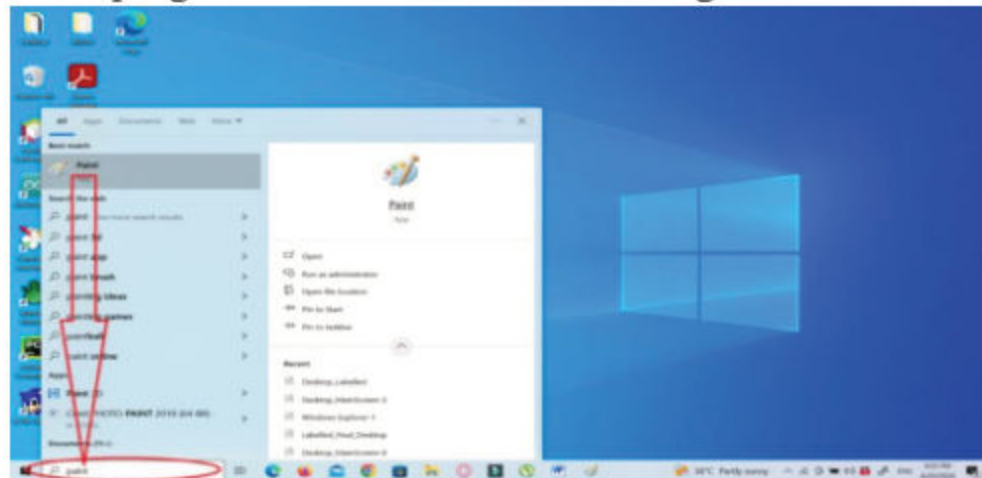


Figure 2.7 Open a file or run an application program from start menu

(iii) Open a file or run an application program from search box

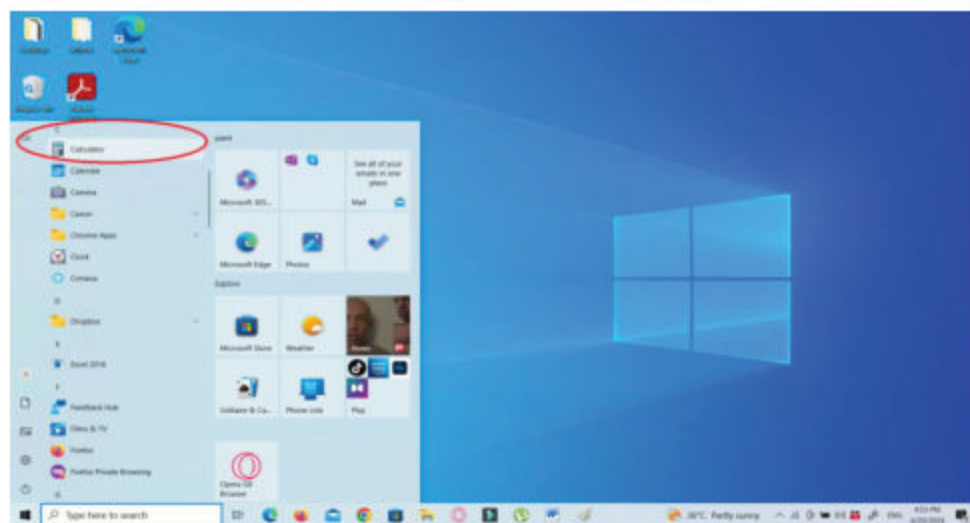


Figure 2.8 Open a file or run an application program from search box

It is very easy and quick way to open a file or run an application from the Windows search box just by writing their names, like type “calc” in search box and press enter key to open a calculator or type “paint” in search box and press enter key to open the MS-Paint program as shown in figure 2.8. Similarly if you type “notepad” in search box and then press enter key, it will open the Notepad text editor program, which is shown in figure 2.9.

ACTIVITY

Opening Programs in Windows 10

Watch and Learn: Your teacher will show you how to open programs like Calculator, Notepad, and Paint.

Hands-On Practice: After the demonstration, you will get to open different programs and files using the start menu, search box, and by double-clicking their icons.

2.5.2 Common Components of Windows Programs or Applications

Every Windows program or application, such as Notepad or MS-Paint, contains some common components, which are shown in figure 2.9.

(i) Title bar:

Title bar is a horizontal bar located at the top of a program window. It displays the program name and the name of opened file. It displays the program name of an opened folder, if you are working in file explorer.

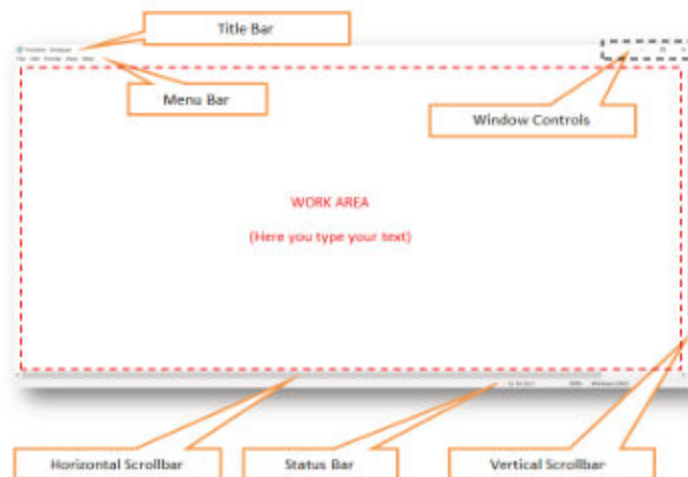


Figure 2.9 Common components of Windows programs (e.g. Notepad)

(ii) Window Controls

The title bar contains three buttons minimize, maximize/restore and close on its top right corner as shown in figure 2.10. These buttons are also known as window controls.

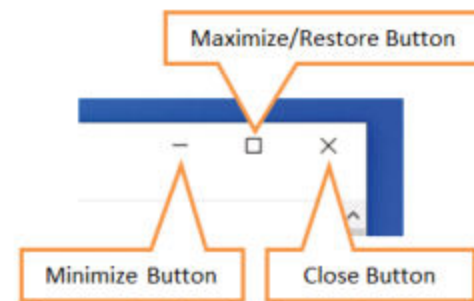


Figure 2.10 Windows Controls

Minimize Button: When you click on minimize button, it reduces/minimizes the current window to task bar. The shortcut key for the minimize button is Windows Key + M.

Maximize/Restore Button: Click on maximize/restore button to enlarge the window to its maximum size, and if the window is already at its maximum size, it switches back to its previous size. The shortcut key for the maximum/restore button is Windows Key + UP Arrow Key.



Note

The maximize/restore button is disabled on programs that are fixed sized programs.

Close Button: When you click on close button, it will close the current program or file. The shortcut key for the close button is ALT Key + F4.

(iii) Menu bar:

A menu bar is a horizontal menu displayed just below the title bar as shown in figure 2.11. It contains a set of menus, like File, Edit, View, etc. When you click any of these menu items, a drop-down menu appears that will show you further options like, New, Open Save, etc.

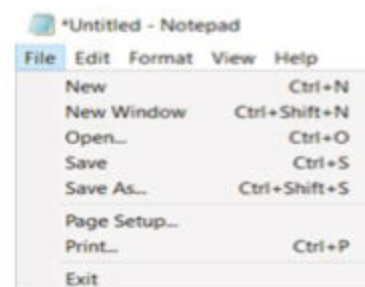


Figure 2.11 Menu bar of Windows "Notepad" program

(iv) Scrollbars:

Scrollbars are windows control elements that appear either on bottom (i.e. horizontal scrollbar) or on the right side (i.e. vertical scrollbar) of a window. Scrollbar are used to scroll left/right or up/down.

(v) Status bar:

Status bar is the horizontal bar at the bottom of the application window showing some status information about the current opened file/document like number of lines and number of characters. When you open file explorer and observe its status bar, you will see that file explorer's status bar displays the information about the number of items (files and folders) available inside it.

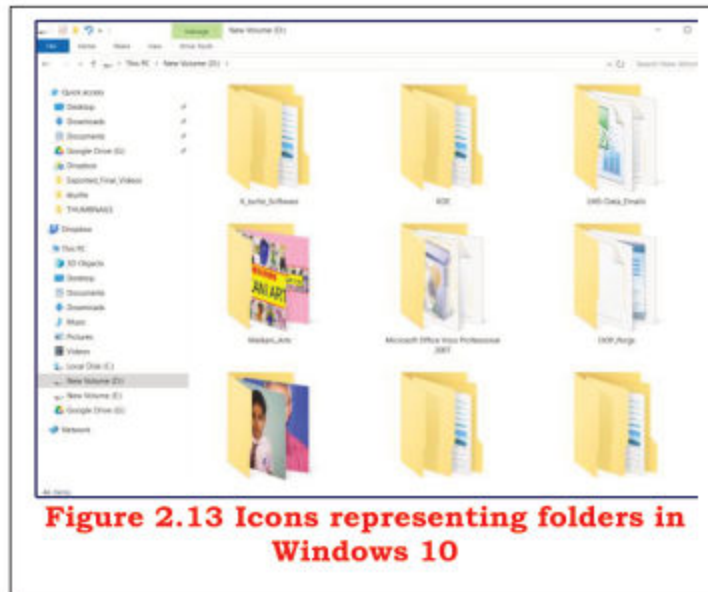
2.6 MANAGING FILES AND FOLDERS**2.6.1 Definition of file, folder, and drive**

A file is an object that stores data and information. It is identified by its given name on a computer. In contrast to file, a folder is a container used to hold different types of files and sub-folders. Folders help in organizing and managing files.

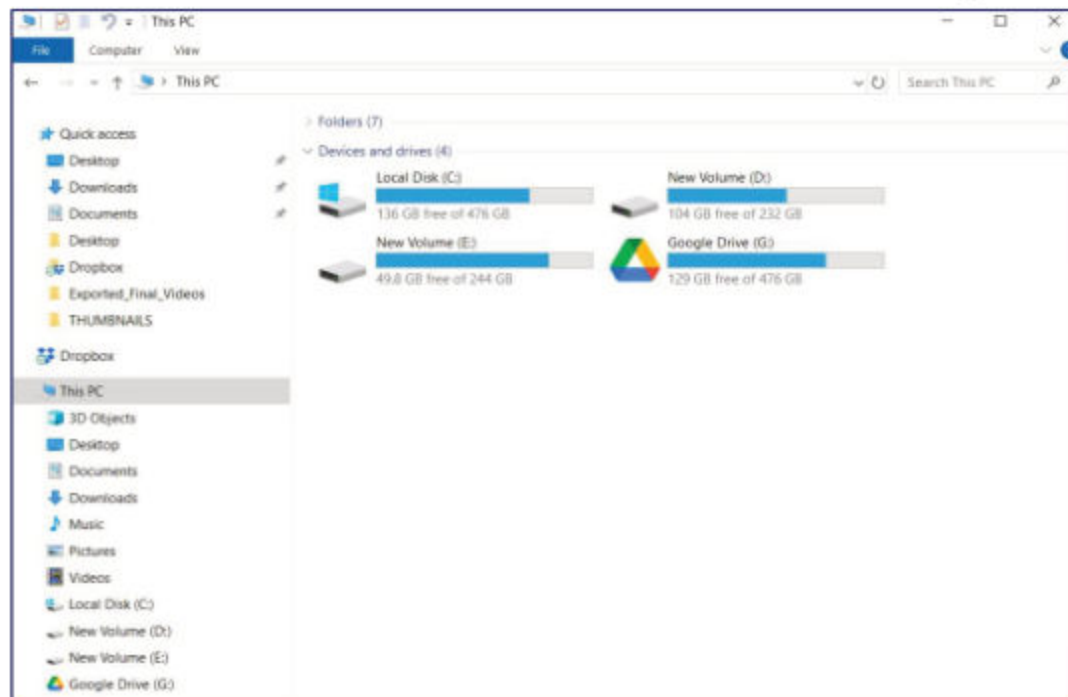


Figure 2.12 Icons of some common types of files in Windows 10

A file can be a text document, picture, audio or video, etc. In Windows, every file type is represented by a separate unique icon. The figure 2.12 shows the icons of some commonly used files, while folder icons are shown in figure 2.13. Files and folders are stored in permanent storage, typically a hard disk.



The hard disk can be divided into section called drives. Each drive is assigned a letter like C, D, E, etc. C drive is usually primary or root drive. When you click on “This PC” icon, it will show these drives and some removable or cloud drives as shown in figure 2.14.



2.6.2 Creating a new file

Follow the below steps to create a new file in Windows 10. All the steps are also depicted in figures 2.15 to 2.18.



Figure 2.15 Create a new file in Windows 10 – steps 1 to 5

1. Click on the folder icon on the taskbar at the bottom of the screen to open the file explorer. Alternatively, you can also open file explorer by pressing the Windows Key + E on the keyboard.
2. Open File Explorer and go to the location where you want to create a new file. Such as the desktop as a folder like my document.
3. Once you navigate to the desired location in File Explore, right-click on an empty space that will open a pop-up menu.
4. In pop-up menu, choose "New" that will show you another sub-menu.
5. Finally, click on "Text Document" in sub-menu that will create a new empty text file.
6. Once file is created, it is automatically named as "New Text Document.txt" as shown in figure 2.16. In case you want to give a different name, then just click on the file name, and type file name of your own choice and press the enter key to save the new file name.



Figure 2.16 Naming newly created file

7. Now, you can open the newly created empty file in the "Notepad" program by double-clicking on its file name/icon.
8. Once file is opened in notepad, you can type your desired text or content into the file as illustrated in figure 2.17.



Figure 2.17 Opening newly created file and typing text into it

9. Once you have finished typing and editing the text, click on "File" menu in the top-left corner of notepad program, and

then click on "Save" as shown in figure 2.18. This will permanently save all of your modifications in file before closing the notepad program.

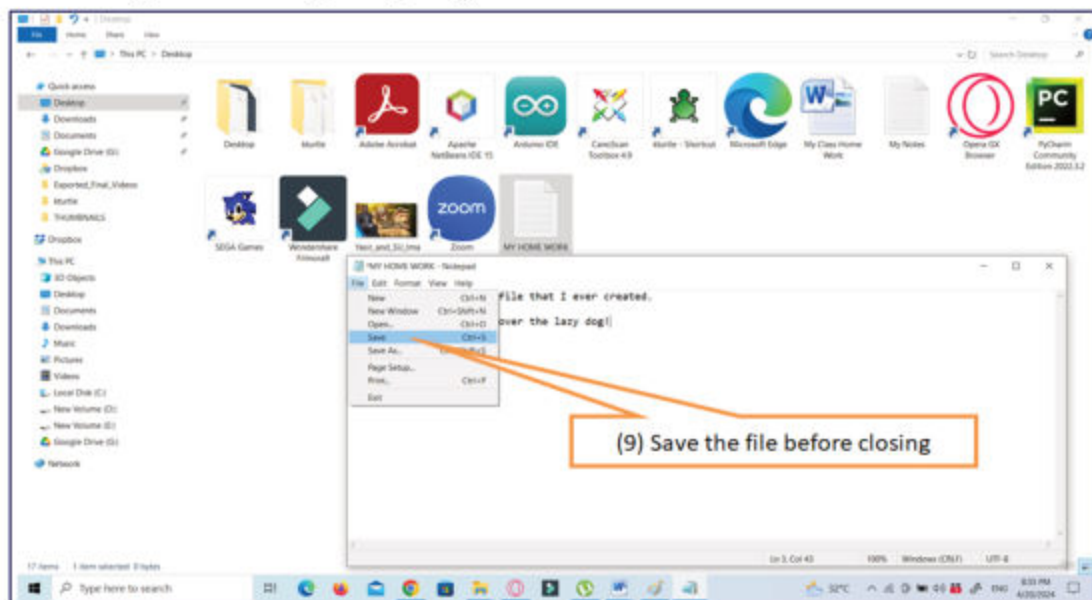


Figure 2.18 Save the file before closing the notepad program

2.6.3 Creating a new folder

The steps to create a new folder are shown in figures 2.19 to 2.21. Note that to create a new folder in Windows 10, the first four steps are exactly same as described in previous section for creating a new file.

1. Open the file explorer.
2. Navigate to the desired location where you want to create the file (in this case a folder).
3. Right click on empty space in file explorer.
4. Choose "New" in pop-up menu that will show you another sub-menu.
5. In the sub-menu, click on "Folder" option that will create a new folder in the current location. Steps 1 to 5 are illustrated in figure 2.19.

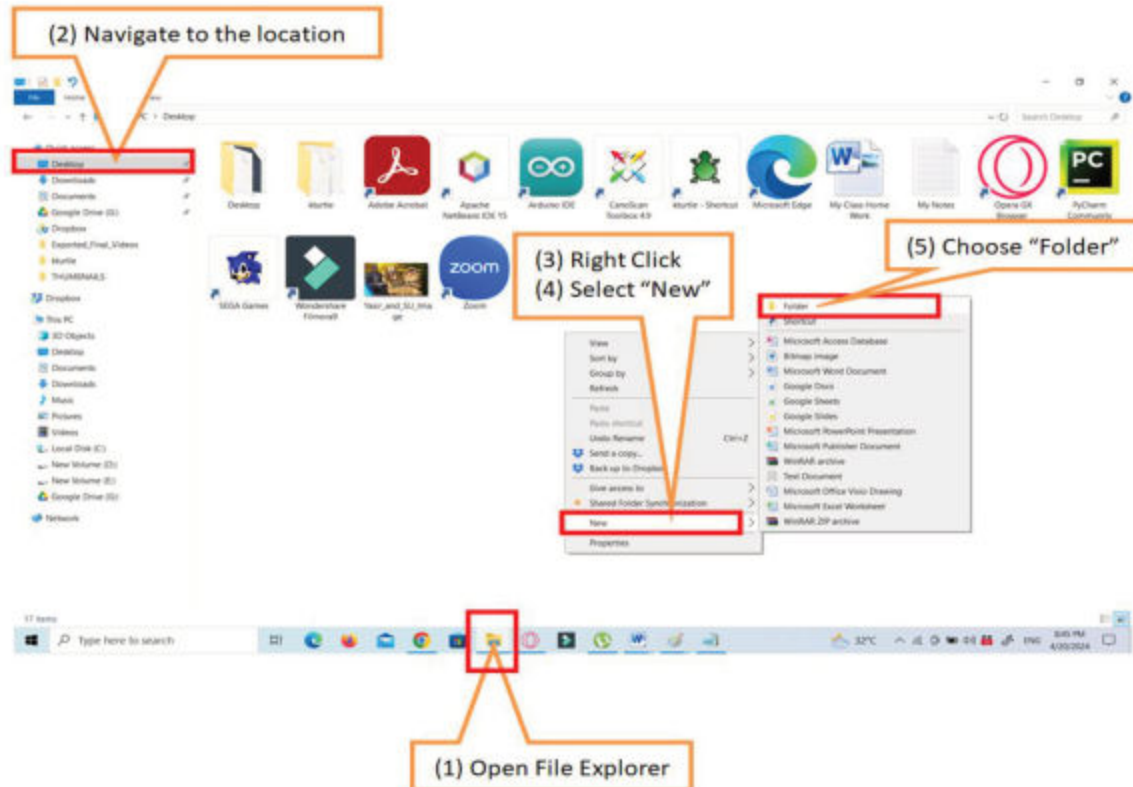


Figure 2.19 Create a new folder in Windows 10 – steps 1 to 5

6. Give name to the newly created folder. The default name is “New Folder”.

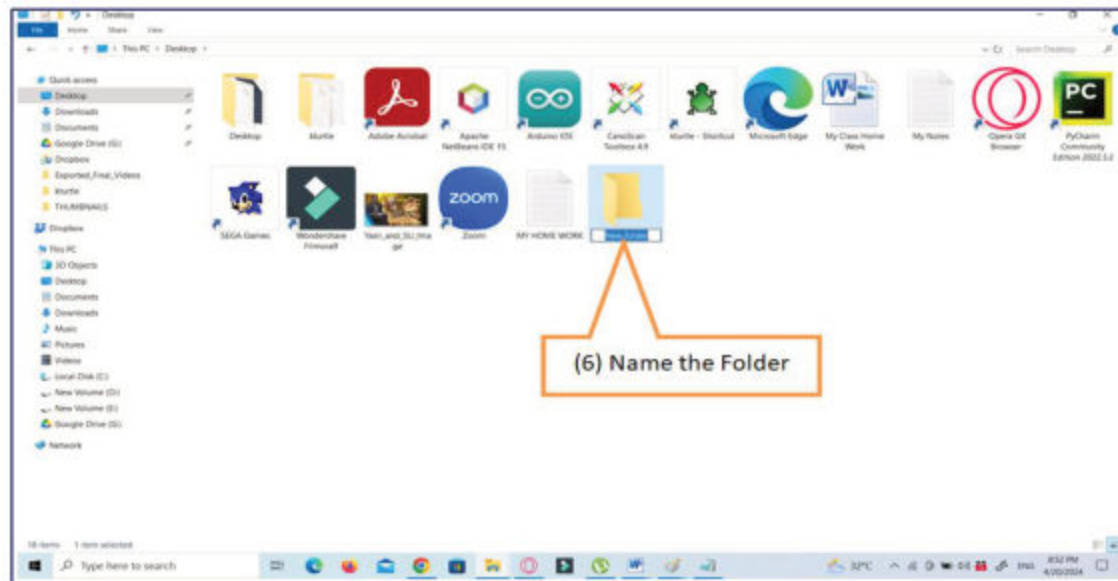


Figure 2.20 Naming newly created folder

7. Double click on the newly created folder to open it. This will show you the contents of the folder as shown in figure 2.21, which is currently empty since you just created it.



Figure 2.21 Newly created empty folder

2.6.4 Renaming a file or folder

The steps to rename a file or folder in windows 10 are listed below, which are also illustrated in figure 2.22.

1. Open the file explorer
2. Navigate to the desired location where you want to rename a file or folder.
3. Single click on the file or folder that you want to rename to select it, and then right click on it, a pop-up menu will appear.
4. Choose “Rename” option in pop-up menu and left click on it, which will make the file or folder name editable. Change the file or folder name by typing a new name, and then press enter to ensure it is renamed.

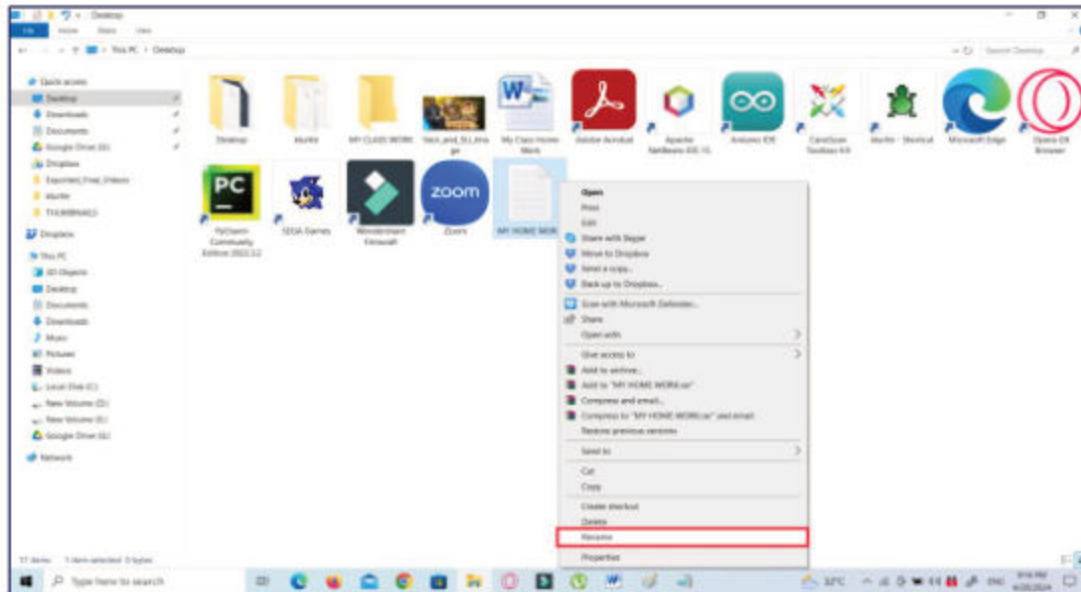


Figure 2.22 Renaming a file (or folder) in Windows 10

2.6.5 Deleting a file or folder

The steps to delete a file or folder in windows 10 are listed below, which are also illustrated in figure 2.23.

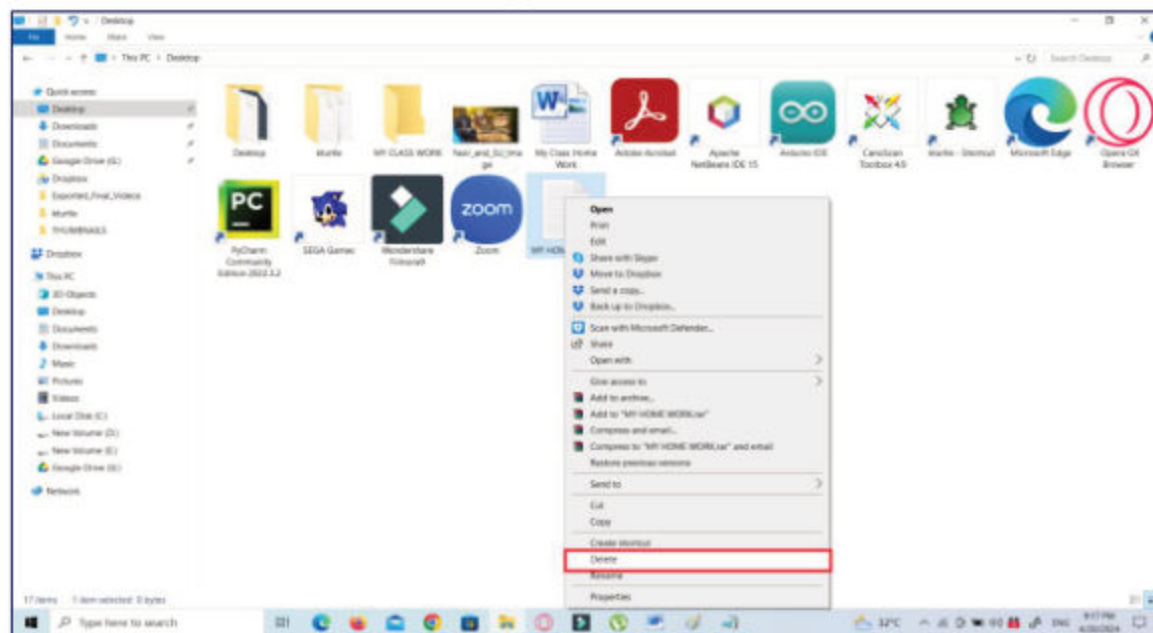


Figure 2.23 Deleting a file (or folder) in Windows 10

1. Open the file explorer and navigate to the desired location from where you want to delete a file or folder.

2. Single click on the file or folder that you want to delete to select it, and then right click on it, a pop-up menu will appear.
3. Choose “Delete” option in pop-up menu and left click on it, which will show you a confirmation dialog box, asking if you are sure to delete the file or folder and move it to the recycle bin. Click on the “Yes” button to confirm that will delete the selected file or folder.

 Note

You can also delete any file or folder just by selecting the file or folder and then press the DELETE button on your keyboard.

 ACTIVITY
Creating and Managing Files and Folders

Watch and Learn: Your teacher will show you how to create a new file or folder, and how to rename or delete them.

Hands-On Practice: After the demonstration, you will:

- Create a text file and a folder.
- Rename the file to “My Class Work”.
- Delete both the newly created file and folder.

2.6.6 Copying a file or folder from one location to another location

Follow the below steps to copy a file or folder from one location to another location in windows 10. All the steps are illustrated in figures 2.24 and 2.25.

1. Open the file explorer.
2. Navigate to the desired source location from where you want to copy a file or folder.
3. Single click on the file or folder that you want to copy to select it, and then right click on it, a pop-up menu will appear.

 Note

You can also copy any file or folder just by selecting the file or folder and then press CTRL+C shortcut key on your keyboard.

-
- (1) Open File Explorer
- (2) Navigate to the source location
- (3) Select the File (or Folder) and Right Click on it
- (4) Copy the File (or Folder)

to the desired destination location.

- [illegible]

6. Choose “Paste” option in pop-up menu and left click on it, which will create a copy of your selected file or folder in this location.

Note

You can also paste any file or folder by pressing CTRL+V shortcut key on your keyboard.

2.6.7 Moving file or folder from one location to another using Cut and Paste options

Follow the below steps to move a file or folder from one location to another location in windows 10. All the steps are illustrated in figures 2.26 and 2.27.

1. Open the file explorer.
2. Navigate to the desired source location from where you want to move a file or folder.
3. Single click on the file or folder that you want to move to select it, and then right click on it, a pop-up menu will appear.
4. Choose “Cut” option in pop-up menu and left click on it.

Note

You can also cut any file or folder just by selecting the file or folder name and then press CTRL+X shortcut key on your keyboard.

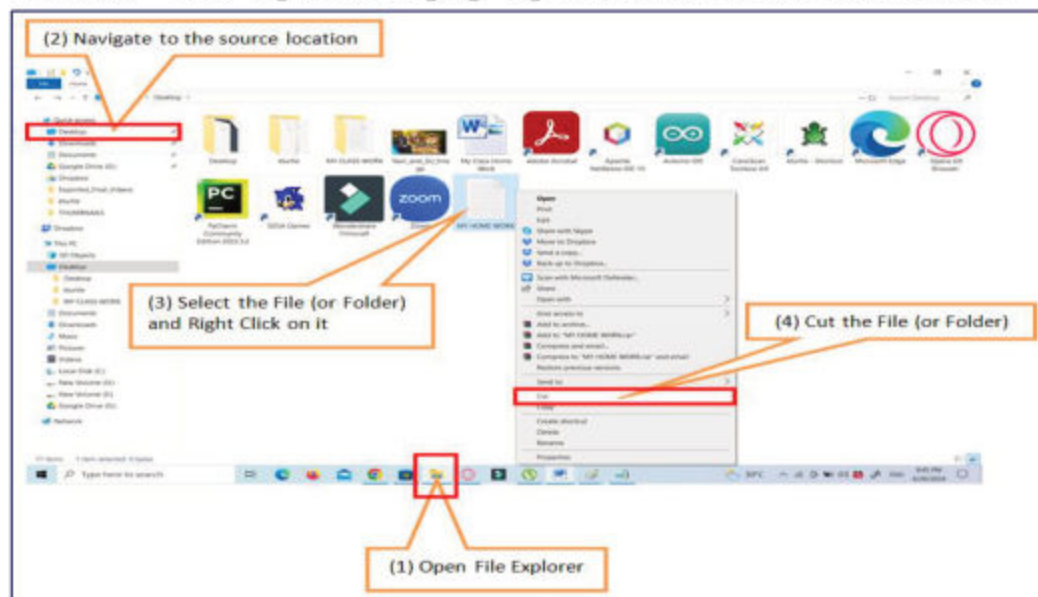


Figure 2.26 Cut a file or folder to move it to another location

5. Navigate to the desired destination location, using the file explorer, where you want to move the file or folder, and then right click on any empty space in the destination location, a pop-up menu will appear.

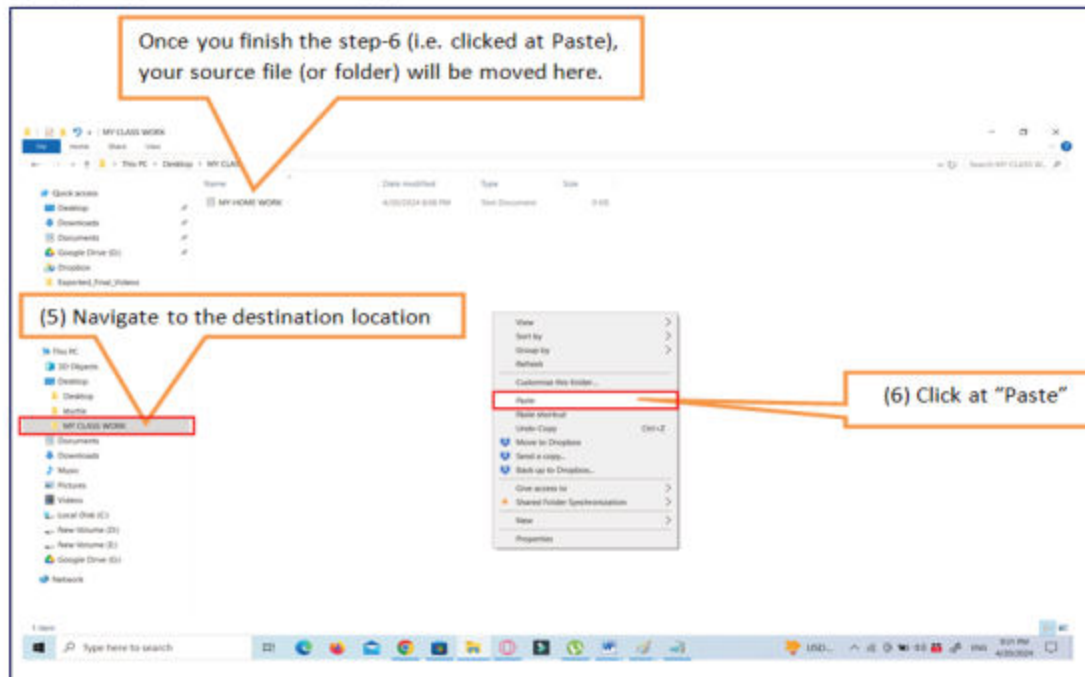


Figure 2.27 Pasting a file or folder to move it in current location

6. Choose "Paste" option in pop-up menu and left click on it, which will move your selected file or folder from source location to this location.

2.6.8 Moving file or folder from one location to another using drag and drop method

Follow the below steps to move a file or folder from one location to another location in windows 10 using a drag and drop method. Open the file explorer.

1. Navigate to the desired source location from where you want to move a file or folder.
2. Single click on the file or folder that you want to move to select it. The selected file or folder will be highlighted.
3. Click and hold the highlighted file or folder, drag it to the desired location, then release the mouse button to drop it. You

will observe that the selected file or folder has been moved from the source location to the destination location.

ACTIVITY

Copying, Cutting, and Pasting Files and Folders

Watch and Learn: Your teacher will show you how to copy, cut, and paste files and folders in Windows 10.

Hands-On Practice: After the demonstration, you will:

- Copy and paste files from one folder to another.
- Move files from one folder to another using the drag and drop method.

2.6.9 Creating a shortcut of file/folder or program

Follow the below steps to create a shortcut of file/folder or program in windows 10. All the steps are illustrated in figures 2.28 to 2.30.

1. Open the file explorer.
2. Navigate to the location of file/folder or program for which you want to create a shortcut, and then single click on it to select it for creating its shortcut.
3. Right click on the selected file or folder, a pop-up menu will appear.
4. Choose "Create Shortcut" from the pop-up menu to create its shortcut in the source location as illustrated in figures 2.28 and 2.29.

OR

To create a shortcut on the desktop, go to the pop-up menu, hover over "Send to" option that will show you another sub-menu. In that sub-menu, click on "Desktop (create shortcut)" option. This will create a shortcut of the selected file/folder or program on Windows 10 desktop. This process is shown in figure 2.30.

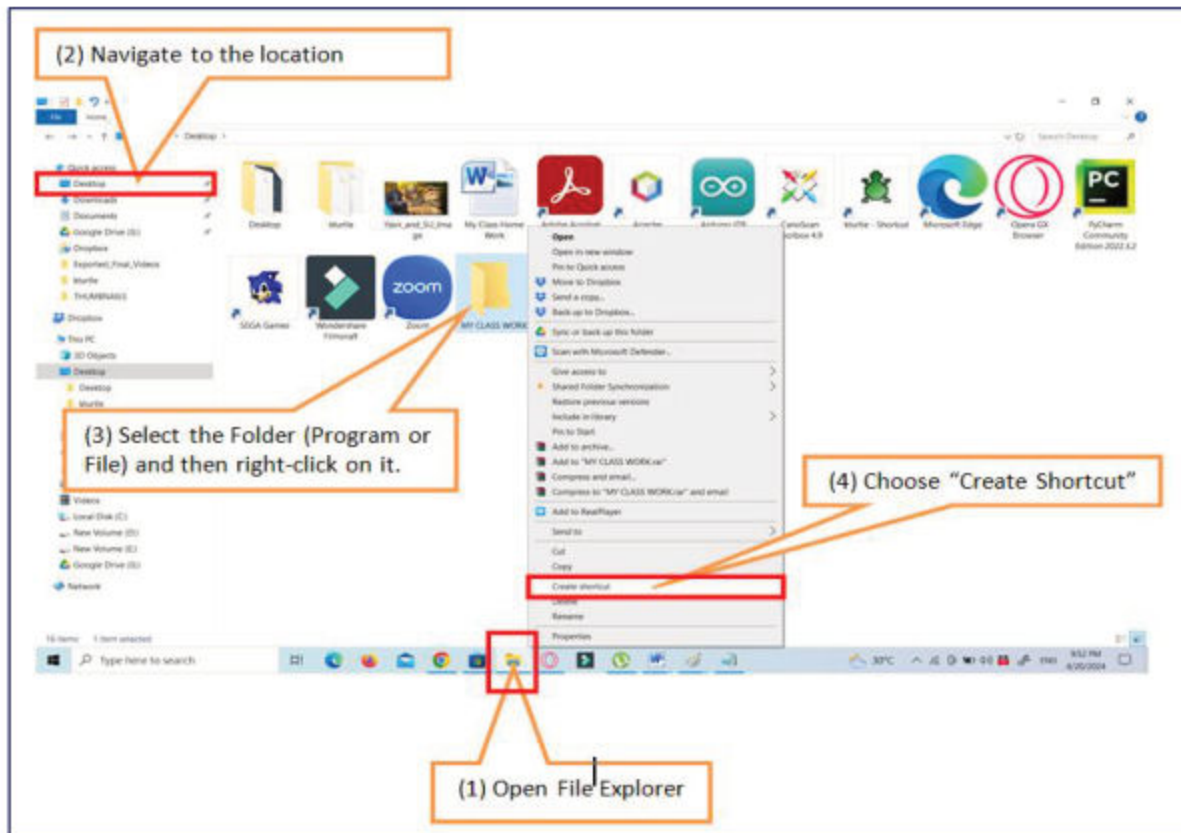


Figure 2.28 Creating shortcut of a file/folder or program

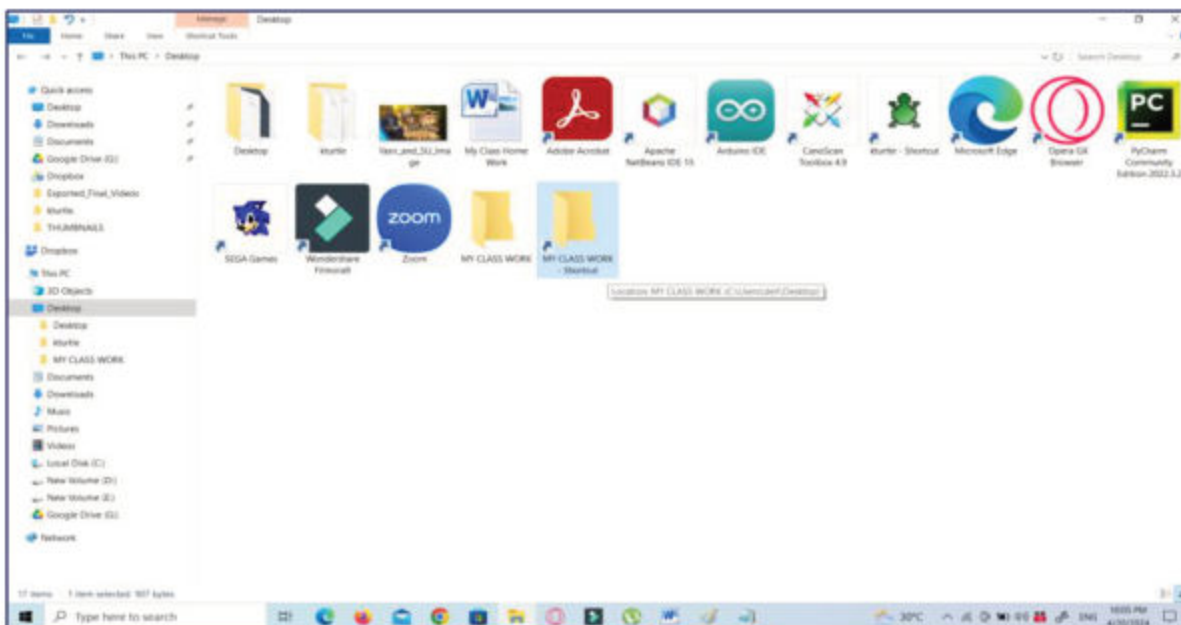


Figure 2.29 Shortcut of selected folder is created in the source location



Figure 2.30 Shortcut of selected folder will be created on desktop

ACTIVITY

Creating Shortcuts

Watch and Learn: Your teacher will show you how to create shortcuts for programs on the Windows 10 desktop.

Hands-On Practice: After the demonstration, you will create shortcuts for your favorite or most frequently used programs on your desktop.

2.7 WORKING WITH MICROSOFT PAINT

Microsoft (MS) Paint is a drawing and simple image processing tool included in all versions of Microsoft Windows operating system.

To open MS Paint in Windows 10, click the start button at the bottom-left corner of the desktop then select the "Paint" option. Alternatively, type "Paint" in the search box next to the Start button on the task bar and press Enter Key or click the "Paint" icon as (see

figure 2.31). This will search for the MS Paint program on your computer, and open it for you.

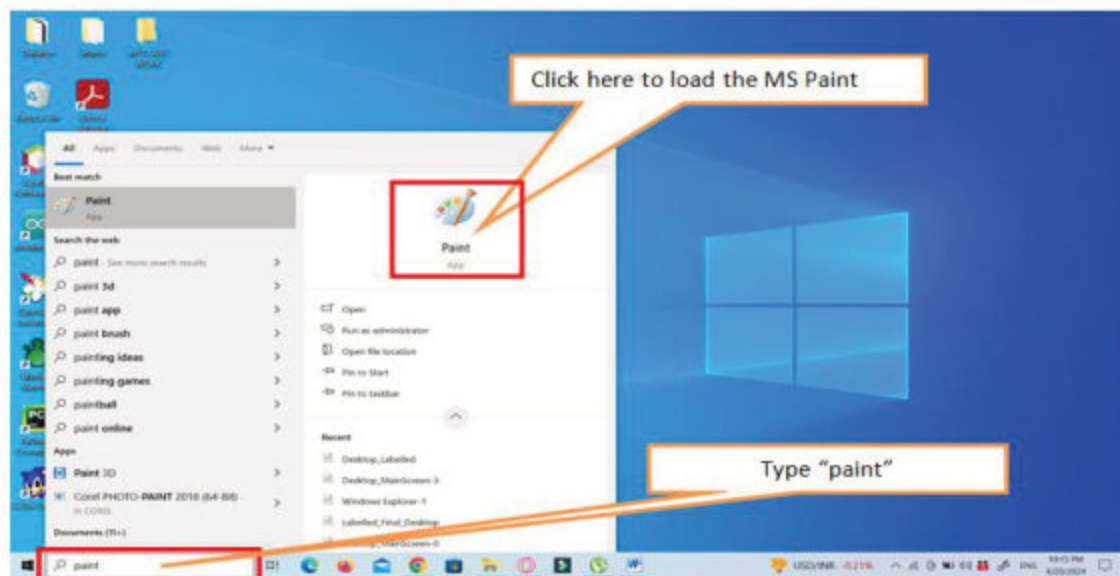


Figure 2.31 Opening MS Paint using the windows search box

2.7.1 MS Paint components and function

Figure 2.32 shows different components of MS Paint program. These components provide you various image processing features like creating freehand drawings, adding shapes, filling colours and adding text to your drawings, etc.

Different components of MS Paint program are described below:

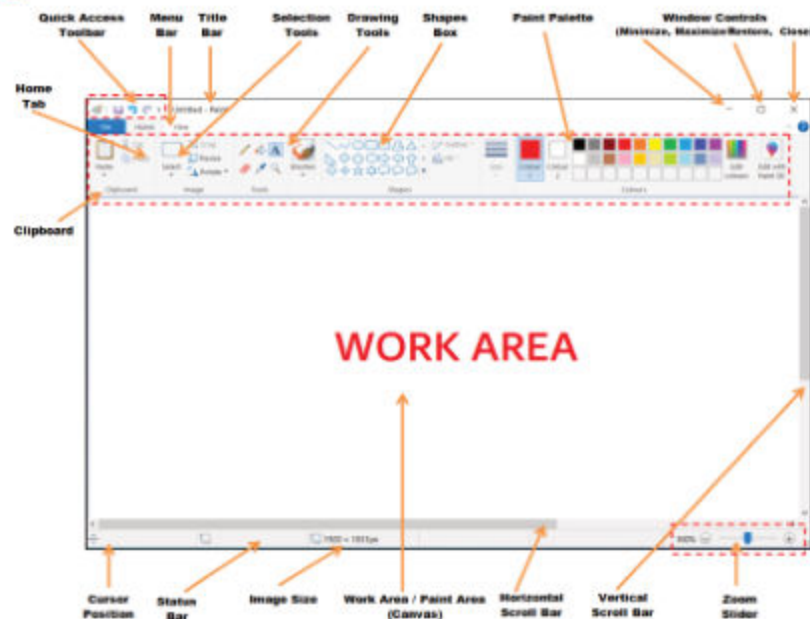


Figure 2.32 Components of MS Paint

Title Bar	Title bar shows the name of the file you are working on and also contains window controls (i.e. minimize, maximize/restore, close buttons) that enables you to adjust the window size of MS Paint as per your needs.
Quick Access Toolbar	Quick Access Toolbar is at the top-left corner of the MS Paint window. It provides quick access to commonly used MS Paint features like Save, Undo, and Print. You can add or remove features of Quick Access Toolbar, because it is customizable toolbar.
Menu Bar	Menu bar gives you access to various menus such as File, Home, and View that ultimately provide you various options and commands for image processing.
Drawing Tools or Tool Palette	Drawing tools or Tool Palette includes drawing and editing tools such as the pencil, brush, eraser, fill, text, colour picker, and magnifier.
Paint Palette	Displays a grid of colours for choosing the drawing and fill colours
Colour Box	Shows the currently selected colour and allows users to pick colours for drawing or filling shapes.
Shapes Box	Contains various drawing tools for creating shapes like lines, squares, circles, rectangles, triangles, polygons, etc.
Paint/Working Area (Canvas)	It is the main area or canvas where you can draw, paint, and edit or process images.
Selection Tools	Selections tools allow you to select and process some particular areas of your drawing/image. You have options for selection tools like rectangular selection, free-form selection, and select all.

Zoom Controls	Zoom Controls are located at the bottom right corner of MS Paint window or can be accessed through the view menu. Zoom controls allow you to zoom in and out of the image for precise and more accurate editing.
Rulers and Gridlines	Rulers and gridlines features can be accessed by clicking at “View” menu item from menu bar. These features assist with aligning and positioning objects in the image accurately.
Status Bar	Located at the bottom of the window, it displays information about the current state of the program, such as the size of the image or the position of the mouse cursor. You can turn it on or off by clicking at “View” menu item from menu bar, and then check or uncheck status bar check box.
Full Screen	You can view your drawing in full screen mode, by clicking at “View” menu item from menu bar, and then click at “Full Screen” option in view tab.

“File” menu of MS Paint contains several submenu items. These items allow you to create new file, open, save, print, and manage images within MS Paint program. Functionality of some of the frequently used items of “File” menu is given below:

Menu Item	Function
New	It allows you to create a new blank image in MS Paint.
Open	It opens an existing image file from the computer's storage in order to view or edit it.
Save	It saves or stores the current image to the computer's storage with its current file name and format.

Save As	In contrast to Save, it allows you to save the current image to the computer's storage with a new file name or format.
Print	It opens the print dialog box in order to print the current image on a connected installed printer.
Set as Desktop Background	It sets the current image as the desktop background wallpaper on your computer.
About Paint	It shows information about MS Paint like version number and copyright details.
Exit	It closes the MS Paint program.

Functionality of various items available by clicking on “Home” tab of MS Paint is given below:

Item/Sub-item Name		Function
Clipboard	Cut	Removes the selected portion of the image and stores it in the clipboard for pasting elsewhere.
	Copy	Copies the selected portion of the image to the clipboard for pasting elsewhere.
	Paste	Inserts the contents of the clipboard into the image at the current cursor position.
Image	Select	Allows users to select a portion of the image for editing, moving, or copying.
	Crop	Cuts out a selected portion of the image, removing everything outside the selection.
	Resize	Resizes the image by adjusting its dimensions, either by percentage or specific measurements.
	Rotate and Flip	Rotates or flips the image horizontally or vertically to change its orientation.

Tools	Pencil	Allows users to draw freehand lines and shapes with a single colour, similar to using a pencil on paper.
	Fill with Colour	Fills a selected area of the image with a chosen colour, allowing for quick colorization.
	Text	Adds text to the image, allowing users to type and customize text directly on the canvas.
	Eraser	Allows users to remove or erase parts of the image by clicking and dragging over them.
	Colour Picker	Enables users to select a colour from the image by clicking on it with the eyedropper tool.
	Magnifier	Opens the Magnifier tool, which magnifies a portion of the image for closer examination and editing.
	Brushes	Offers various brush styles and sizes for freehand drawing and painting on the canvas.
Shapes		Inserts predefined shapes such as lines, rectangles, circles, and polygons into the image.
Size		It allows users to adjust the thickness or size of the lines or shapes they draw with certain tools, such as the pencil, brush, or line tool.
Colour1		This box represents the primary colour. When users click on Colour 1, they can choose a colour from the colour palette to set it as the primary colour for their drawing tools. The primary colour is typically used for drawing lines, shapes, and outlines.
Colour2		This box represents the secondary colour. Users can also choose a colour from the colour palette and set it as the secondary colour. The secondary colour is often used for filling shapes or adding colour accents to the drawing.
Colours Palette		Enables users to select a colour from the colour palette to use for drawing or filling shapes.

2.7.2 Creating a new file and saving it in MS Paint

When you open MS Paint in Window 10, it automatically opens a new blank file for you. In case you have already opened MS Paint and now you want to create a new file and save it, then follow the below steps:

1. Click at the "File" menu at the top-left corner of the MS Paint window that will show you a dropdown menu.
2. By clicking on "New" in the dropdown menu, as shown in figure 2.33, a new blank MS Paint file will be created.

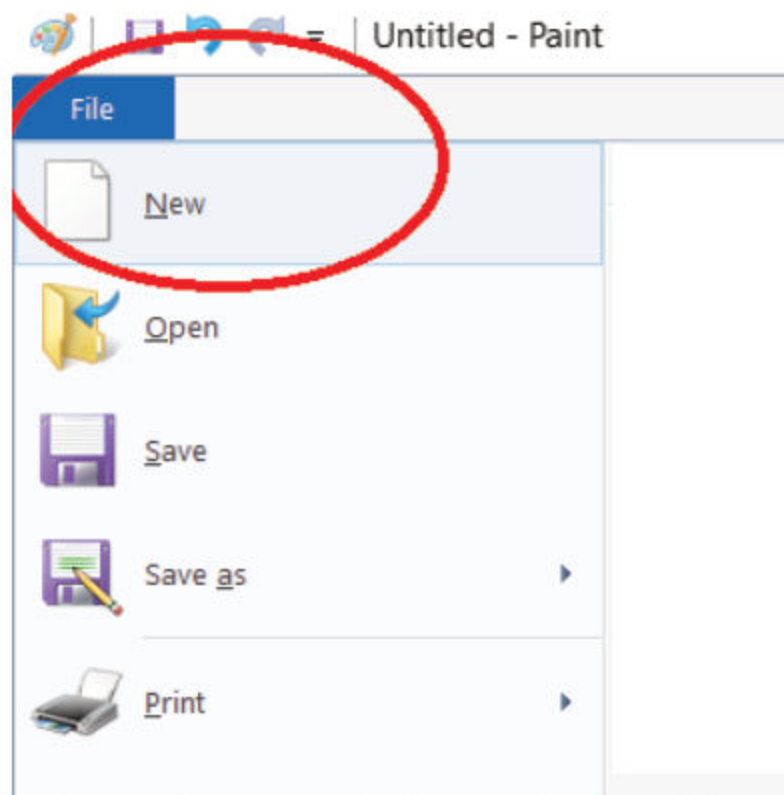


Figure 2.33 Creating a new file in MS Paint for Window 10

3. Once your new file is created, you can start drawing or painting (as shown in figure 2.34) using the various tools and colours available in MS Paint, which are described in previous section.
4. After you have finished creating or editing your drawing, you can save it by clicking on the "File" menu, which will show

you the dropdown menu. Click at “Save” option in dropdown menu as shown in figure 2.35(a), it will show you the save dialog box. Choose your desired location on your computer to save or store the file, and then give it a name and choose image file type as shown in figure 2.35(b). Finally, click at "Save" button in save dialog box.



Figure 2.34 Creating a drawing in MS Paint

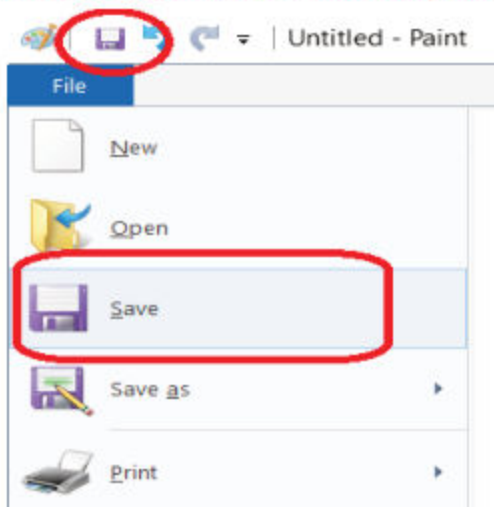


Figure 2.35(a) Saving a file in MS Paint part-1

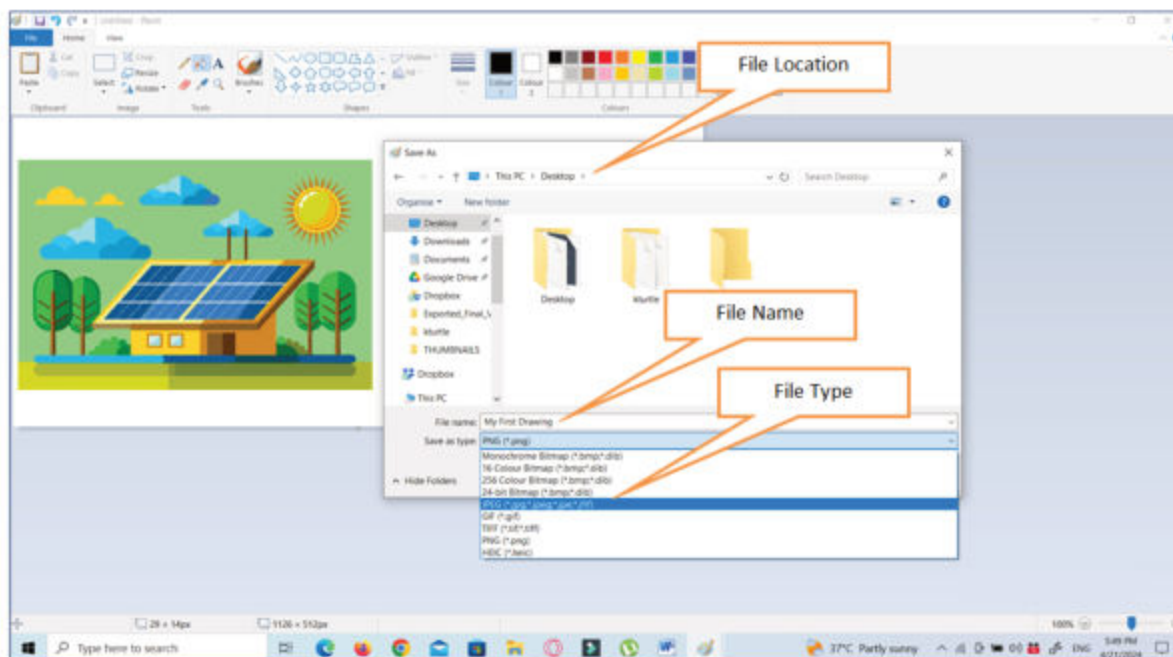


Figure 2.35(b) Saving a file in MS Paint part-2

2.7.3 Opening an existing image file for editing in MS Paint

Follow the below steps to open an existing image file in order to view or modify it in MS Paint:

1. Click at the "File" menu at the top-left corner of the MS Paint window that will show you a dropdown menu.
2. Click on "Open" option in the dropdown menu (as shown in figure 2.36), it will show you the open dialog box. Using the open dialog box, navigate to the location where your desired image file is saved/stored. Once you have found the file, click on its file name and then click at "Open" button (as shown in figure

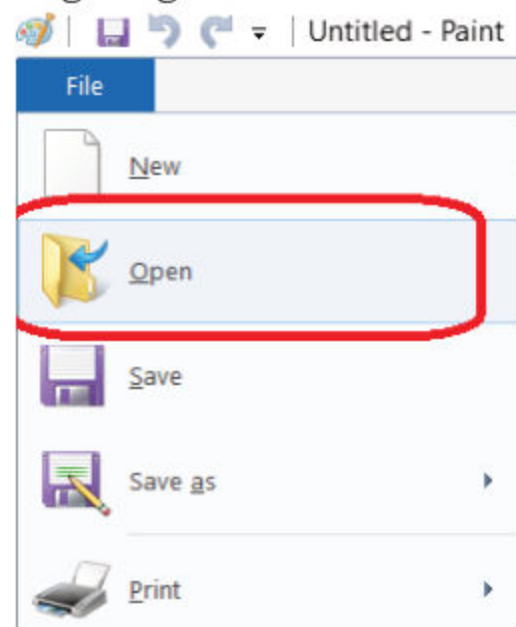


Figure 2.36 Opening an existing image file in MS Paint part-1

2.37) that will open the selected file in MS Paint for viewing and editing purposes.

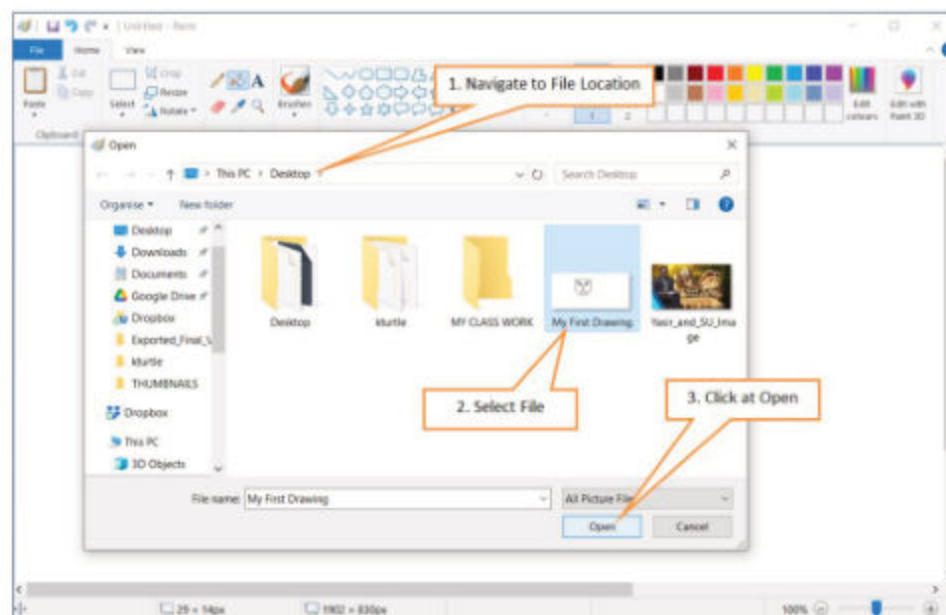


Figure 2.37 Opening an existing image file in MS Paint part-2



Note

You can also double click the file name to open it quickly without clicking at “Open” button.

2.7.4 Creating and editing freehand drawing in MS Paint

Follow the below steps to create and edit freehand drawing in MS Paint:

1. In the toolbox, select the pencil or brush tool by clicking on their icon as per your needs. The figure 2.38 shows a drawing in which pencil tool is used to draw the boat, while water part is drawn using a brush tool.
2. After choosing the pencil or brush tool, select a colour by clicking on one of the colours in the colour palette.
3. Start drawing your desired freehand art in the paint area (canvas). Click and hold the left mouse button, then move the mouse to draw freehand lines on the paint area. Release the mouse button to stop drawing.

4. If you want to adjust the thickness of your lines, you can do so by clicking on the "Size" option located near the colour palette. Choose a size from the dropdown menu that seems appropriate to you for your freehand drawing.
5. If you make a mistake while drawing, you can undo it by clicking on the "Undo" button in the Quick Access Toolbar located at the top-left corner of the window. You can also undo last action by pressing the CTRL + Z shortcut key.
6. To redo something that you have undone, click at "Redo" button in the Quick Access Toolbar located at the top-left corner of the window. You can also reverse/redo the most recent modification made using the "Undo" option by pressing the CTRL + Y shortcut key.
7. If you want to erase part of your drawing, you can select the eraser tool from the toolbox and use it to remove unwanted lines.

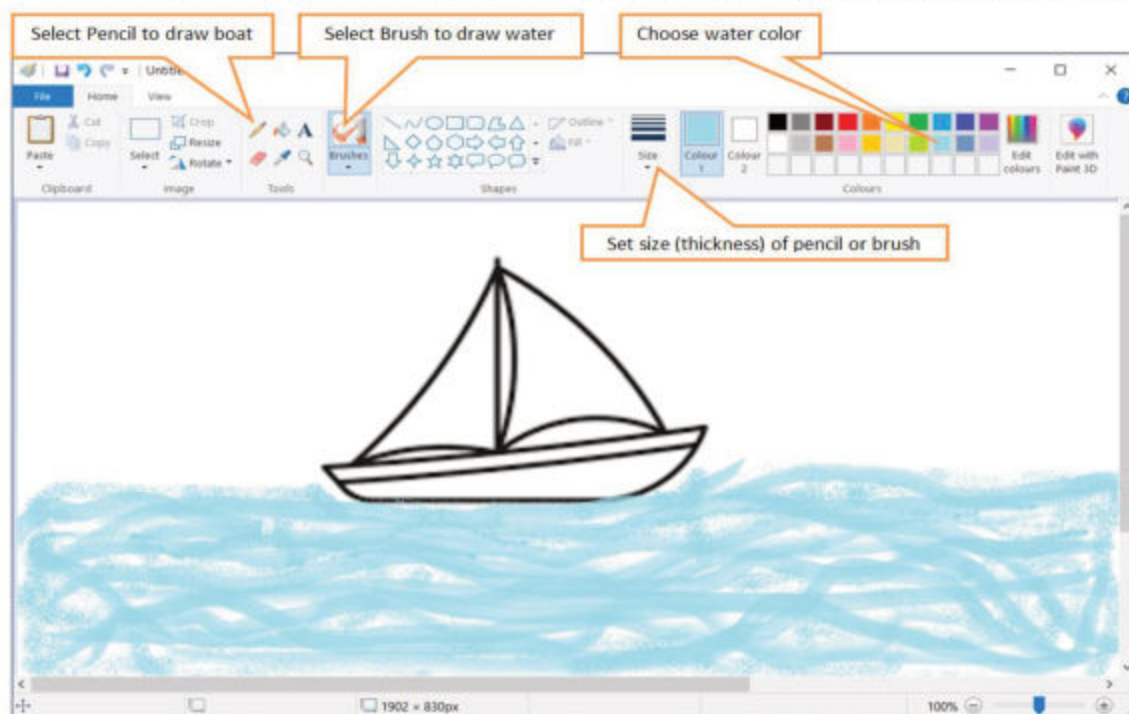


Figure 2.38 Freehand drawing in MS Paint

ACTIVITY

Drawing Your Own Masterpiece

Watch and Learn: Your teacher will show you how to create freehand drawings in MS Paint.

Hands-On Practice: After the demonstration, you will use the skills you've learned to draw the image as shown in figure 2.38 or any other similar image of your own choice.

2.7.5 Adding text to drawings and images

Follow the below steps to add text to your drawings in MS Paint.

1. Click on the "Text" icon in the toolbox to select the text tool for adding text to your drawings in paint area/canvas. After selecting the text tool, move the mouse cursor in paint area at the position where you want to add the text, and click mouse button. This will create a text box for you in which you can type your desired text.
2. Before typing text in the text box, choose the desired font style and text size from the dropdown menu located in the "Font" section of "Text" menu. Also click on the "Colour 1" box located near the colour palette to choose the colour for your text.
3. Once you have set the font style, size and colour, start typing your text using the keyboard as shown in figure 2.39.
4. If you want to change the text size, font, or colour after typing the text, you can do so by selecting the "Text" menu with the



Figure 2.39 Adding text to drawings in MS Paint

mouse, and then adjust the properties using the available options in the toolbar as shown in figure 2.40.

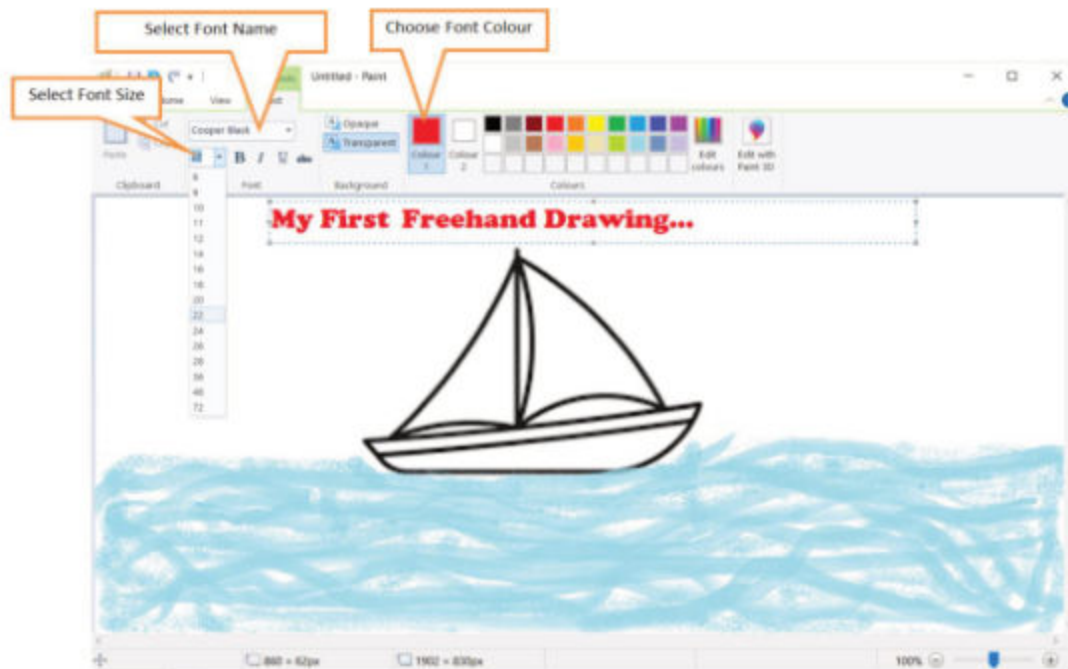


Figure 2.40 Adjusting the properties of text

ACTIVITY

Using Text in Microsoft Paint

Watch and Learn: Your teacher will show you how to add text to drawings and images in Microsoft Paint.

Hands-On Practice: After the demonstration, you will:

- Draw different shapes on the paint canvas.
- Use different colours, font styles, and sizes to add text, like the names and description of the shapes, to each of them.

2.7.6 Move or position the drawings as needed in MS Paint

Follow the below steps to move or position the drawings as per your needs in MS Paint:

1. Click on the image or drawing you want to move to select it. Selection handles (small squares) around the edges of the drawing or image indicates that the image or drawing has been selected.

2. Once the drawing is selected, move your mouse cursor over the drawing until it changes to a cross with four arrows. Click and hold the left mouse button, then drag the drawing to the desired location in the paint area, and release the left mouse button to place it there.

ACTIVITY

Shape Your Imagination

Watch and Learn: Your teacher will show you how to draw shapes, fill them with colour, and use tools like rotate, flip, resize, and skew in Microsoft Paint.

Hands-On Practice: After the demonstration, you will:

- Draw shapes of your choice on the canvas.
- Fill each shape with your favourite colours.
- Experiment with rotating, flipping, resizing, and skewing your shapes to create unique designs.

2.7.7 Steps to outline or fill colours in drawings

Follow the below steps to outline or fill colours in drawings in MS Paint:

1. Click on the shape or drawing you want to outline or fill colour.
2. To outline the drawing, click on the "Shapes" tool in the toolbox. Choose an appropriate shape (e.g. circle, ellipse, rectangle, etc) to outline your drawing.
3. Click and drag on the paint area to draw the shape that will create an outline around the desired drawing.
4. To fill the drawing with colour, click on the "Fill with Colour" tool in the toolbox, and then click inside the drawing to fill it with the currently selected colour from the colour palette.
5. If you want to change the outline colour or fill colour, click on the "Colour 1" or "Colour 2" box near the colour palette to

choose a new colour as per your needs, and then click inside the outline or fill area to apply the new colour.

ACTIVITY

Digital Greeting Cards

Watch and Learn: Your teacher will show you how to design and personalize greeting cards using Microsoft Paint.

Hands-On Practice: After the demonstration, you will:

- Choose an occasion (e.g., birthday, holiday, thank you) for your greeting card.
- Design a custom greeting card with colourful backgrounds, drawings, and text in MS Paint.
- Share your digital greeting cards with your classmates, friends or family members.

2.8 NAVIGATING THE INTERNET

2.8.1 Definition of Network:

Network in computer science (as shown in Figure 2.41) is a group of computers that are connected to each other through wired or wireless medium to share hardware and software resources and exchange data and information among users.

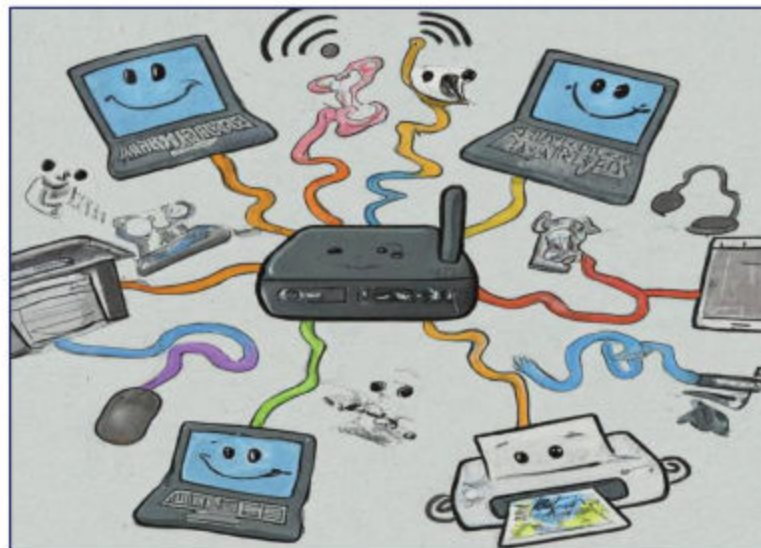


Figure 2.41 Computer Network

2.8.2 Definition of Internet:

The Internet is an international network (also known as global network) consisting of millions of public, private and business networks. It connects people, places, and many other digital resources around the world. Its users can explore, learn and easily connect with other users in order to share information or other interesting things, such as photos, videos, games, tutorials, etc.



Figure 2.42 Scenario of the Internet

2.8.3 Applications of Internet

- **Explore websites:** Visit websites to learn about your favourite topics, play games, watch videos, or read stories.
- **Connect with friends:** Use social media to chat with friends, share photos and videos, or play online games together.
- **Learn something new:** Find information about anything you can imagine, from history and science to cooking and crafts.
- **Online shopping:** You can buy things, such as books, clothes, toys, gadgets, mobile phones, home accessories, grocery items, etc, from online stores without leaving your home.
- **Keep yourself informed/updated:** Watch news channels, read news articles, watch videos on current affairs, or listen to podcasts to stay up-to-date on what's happening in your surroundings and in the world.

2.8.4 Safety precautions while using the Internet

Though the Internet is a very useful and amazing place, but it is essential to stay safe when you are online and using the Internet. You must follow the below guidelines while using the internet to stay safe:

- **Keep your personal data private:** Never share your personal data (such as, your full name, phone number, bank account number, debit/credit card information, address, or passwords) publicly on the internet.
- **Verify web links before you click:** Unknown or non-verifiable web links and downloadable files may contain harmful code, therefore be cautious when clicking on web links or downloading files.
- **Be kind and respectful:** Always treat others online with kindness and respect. This way, others will also treat you nicely and respectfully.
- **Do not be late in talking to a trusted adult:** In case you ever feel uncomfortable while using the internet and interacting with people on social media or gaming platforms or come across something that doesn't seem right online, talk to a parent, teacher, or other trusted adult immediately.

2.8.5 World Wide Web (WWW)

World Wide Web or simply www is a data and information sharing system of interlinked hypertext documents, which are accessed via the internet.

2.8.6 Web Page and Website

A single document shared on the Internet is called a web page, while collection or group of web pages is called a website. These web pages or websites contain information in the form of documents, simple text, images, audio and video files. A web page or website is accessible through the Internet by using a web browser. Every website has its address URL like www.microsoft.com. A website is published and maintained by an organization or an individual.

2.8.7 Web Browser and Their Functionality

A web browser is the software that provides you an interface to access websites on the internet. Currently, different types of web browsers are available to access websites on the internet. Some of the very well-known browsers are listed below:

Microsoft Internet Explorer 	Microsoft Internet Explorer is a web browser developed by Microsoft Corporation. It comes pre-installed with all older versions of Windows operating system.
Microsoft Edge 	Microsoft Edge is a web browser, which is also developed by Microsoft Corporation and it comes pre-installed in Windows 10 and later versions. Microsoft Edge is designed for speed, security, and compatibility with modern web standards.
Google Chrome 	Google Chrome is developed by Google Inc. It is one of the most widely used web browser worldwide. Google chrome is famous for its simplicity, speed, and vast support for web features and standards.
Mozilla Firefox 	Mozilla Firefox is an open-source web browser developed by the Mozilla Foundation. Its main focus is privacy, security, and customization options, which allow users to install add-ons and themes to enhance their browsing experience.
Opera 	Opera is a feature-rich web browser known for its speed, built-in ad blocker, and innovative features such as Opera Turbo, which compresses web pages for faster loading on slow connections, and a free virtual private network (VPN) service, etc.

Safari

Safari is the default web browser for Apple devices that includes iPhones, iPads, and Mac computers.

Safari is recognized for its speed, energy efficiency, and integration with other Apple services and features.

Some of the common functions or activities that can be performed using web browsers are listed below:

- Browsing/surfing websites
- Reading news and articles
- Reading story books
- Research and learning
- Listening to music
- Watching videos and movies
- Online shopping
- Playing games
- Social networking
- Online banking and payments

2.8.8 Opening a website using the web browser

In order to open any website or search engine to search the content of your interest, you need to follow the below steps:

1. Click on the icon of web browser that is available on windows desktop or taskbar (i.e. check it out in pinned apps).
2. Type the complete and correct web address of the website in address bar (e.g. <http://stbb.edu.pk/>) and press enter key.
3. Once you press the enter key, the browser will start searching the website.
4. After a few moments, the home page of the website will be loaded into web browser as illustrated in figure 2.43. You can now explore the website to find out your desired information or services.

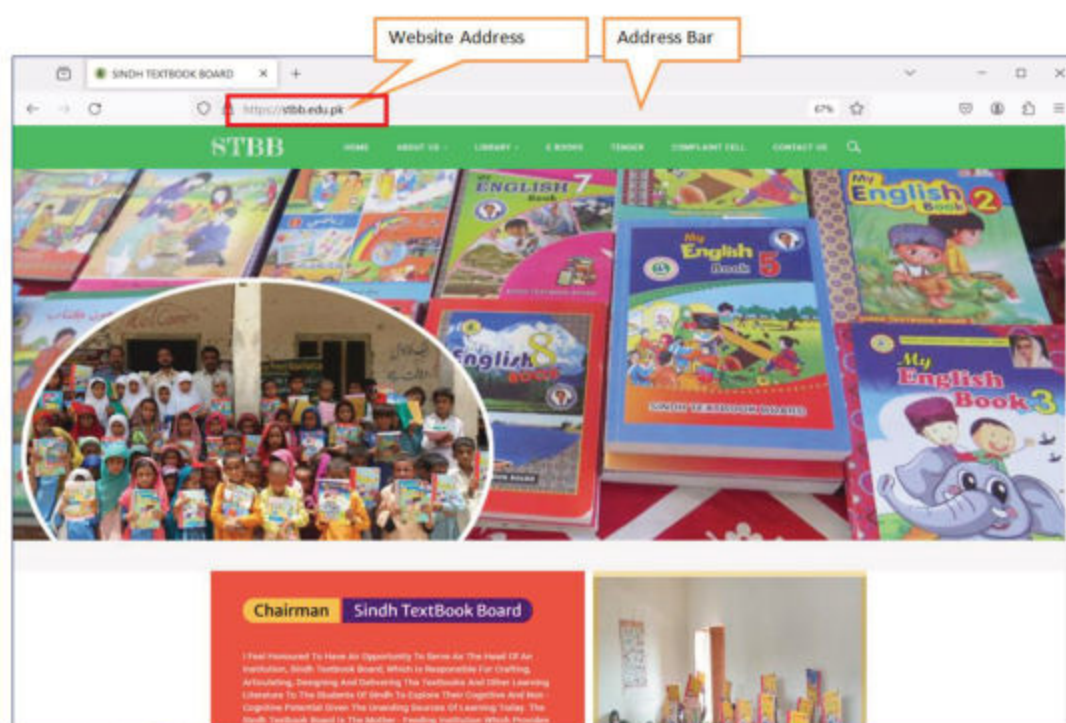


Figure 2.43 Home page of Sindh Text Book Board website

ACTIVITY

Safe Web Browsing

Watch and Learn: Your teacher will show you how to explore websites safely and responsibly.

Hands-On Practice: After the demonstration, you will:

- Browse websites related to your favourite subjects or hobbies.
- Read articles, watch videos, or play interactive games on educational websites.
- Share with the class what you learned or found interesting during web browsing.

2.9 GETTING FAMILIAR WITH SEARCH ENGINES

2.9.1 Search Engine

Search engines are websites that help users to find desired information on the Internet quickly and easily. You just need to type few keywords or simple phrases relevant to the topic or information

you want to search. The search engine intelligently searches the information from other websites based on given keywords or phrases and returns you the list of websites links that contains these keywords or phrases. Some common search engines are shown in figure 2.44.



Figure 2.44 Common search engines

2.9.2 Using search engines to look for certain topics

In order to search a topic (in this case “What is Artificial Intelligence”) using a search engine like Google, follow these steps:

1. Open any available Web Browser, such as Microsoft Edge, Fire Fox or Google Chrome.
2. Type the website address of search engine (in our case www.google.com) in address bar of web browser and press Enter Key.
3. Once Google search engine is loaded in web browser, type your search query in the search bar. For example, type “what is artificial intelligence” and press the “Google Search” button as shown in figure 2.45. After few moments, search engine will display list of search results that are relevant to given search query as shown in figure 2.46. Note that most relevant search results are always appear on top of the list. In case you are not satisfied with the initial search results, you can filter them or refine them by changing the search query.

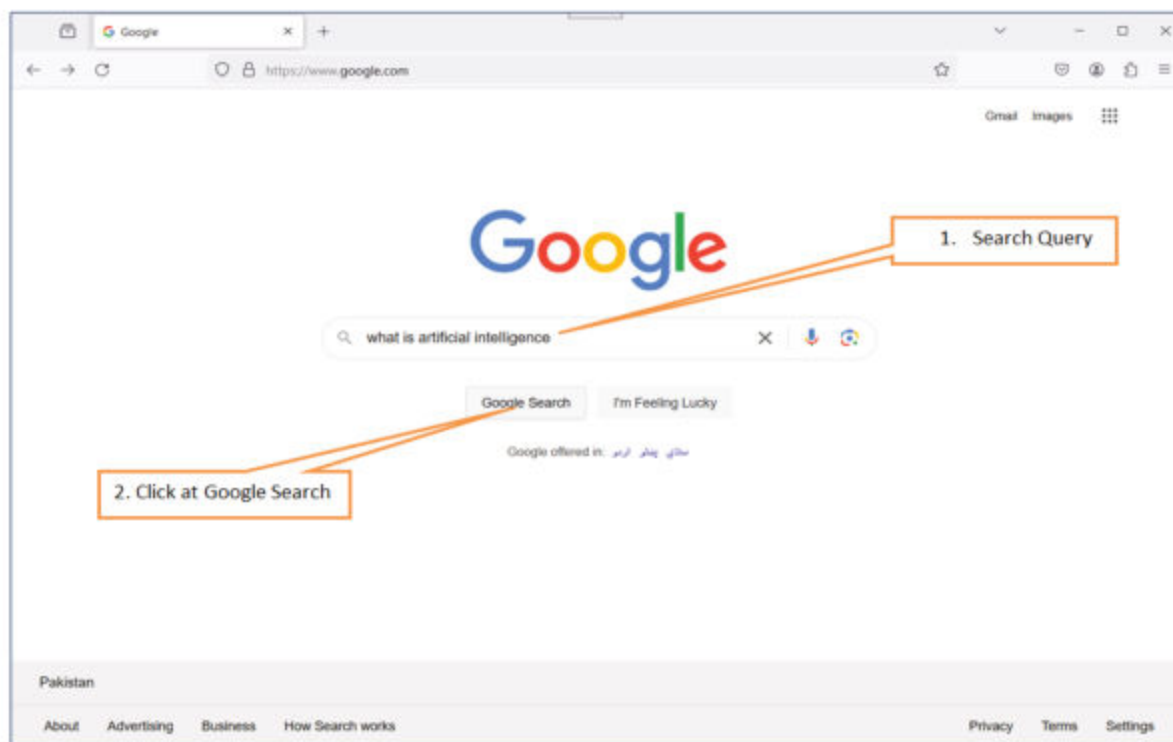


Figure 2.45 Searching for certain topic using Google search engine

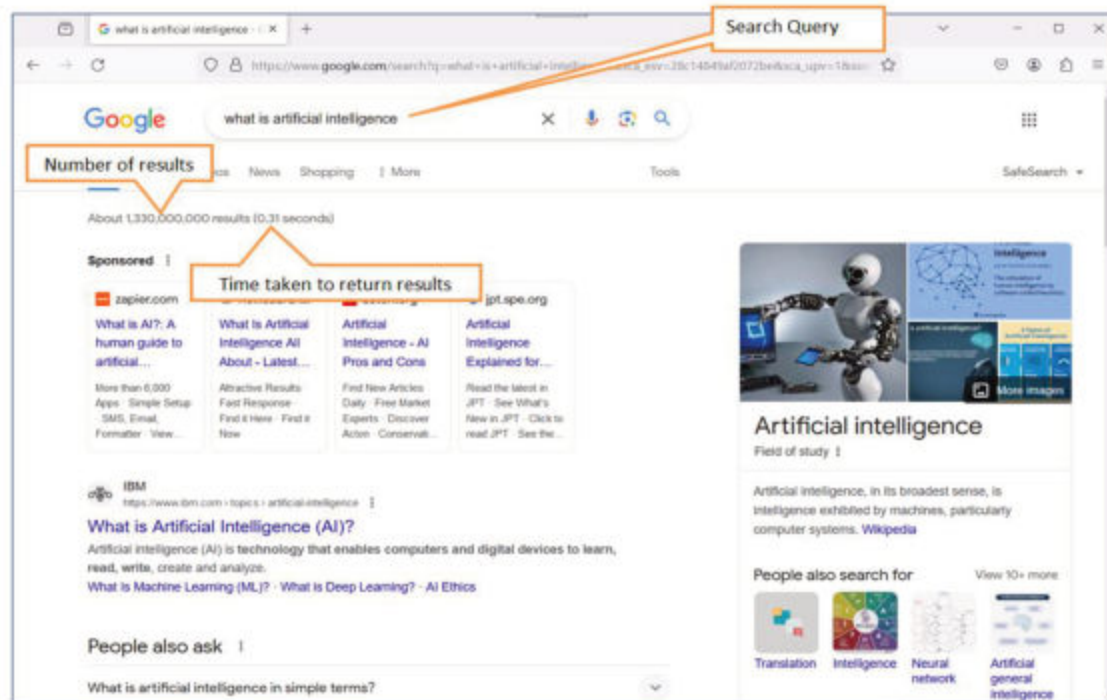


Figure 2.46 Search results returned by Google search engine

 ACTIVITY**Using Search Engines**

Watch and Learn: Your teacher will show you how to search for answers to questions or topics online using Google or other search engines.

Hands-On Practice: After the demonstration, you will search for answers to questions or topics provided by your teacher. For example:

- Look up information about your favourite personality.
- Look up information about an animal and its habitat.
- Search for a recipe for your favourite dish.

Share what you found with your classmates and talk about interesting things that you discovered using search engines.

Summary

- This unit covered basic concepts of the operating system including the functions and working of Windows 10 operating system, image processing using MS Paint, and internet usage.
- An operating system is the system software that monitors and controls computer hardware and software resources, to make it easy for human to use the computer. Different types of operating systems for computers and mobile phones, such as Windows, Mac OS, Linux, Android, iOS, and Windows 10 Mobile, are explained.
- Windows 10 startup procedure is explained along with step-by-step instructions for performing various common tasks, such as opening files or applications, renaming, deleting, copying, moving files or folders, and creating shortcuts. Additionally, common components of Windows programs, such as title bars, window controls, menu bars, scrollbars, and status bars, are explained in detail.
- Microsoft Paint (MS Paint) is a drawing and image processing tool bundled with Microsoft Windows. To access it in Windows 10,

- you can either click on the "Paint" option in the start menu or search for it in the taskbar.
- MS Paint consists of several components such as the title bar, quick access toolbar, menu bar, drawing tools, paint palette, etc. These components offer various functions such as creating freehand drawings, adding shapes, and filling colours, etc.
 - The File menu in MS Paint allows creating, opening, saving, printing, and managing images. The Home tab provides tools for clipboard management, image editing and drawing.
 - To create and save a new file in MS Paint, you can click on "New" in the File menu, draw using the available tools, and then save using the "Save" option.
 - To open an existing image file, you can select "Open" from the File menu and navigate to the desired file and, double click its name/icon.
 - MS Paint allows creating and editing freehand drawings by selecting the pencil or brush tool, choosing a color, and drawing on the canvas.
 - Text can be added using the Text tool, and drawings can be moved using the selection tools.
 - In navigating the Internet, understanding networks and the Internet itself is very important. A network connects computers to share resources, whereas the Internet links millions of networks worldwide, allowing users to explore, learn, and share information like photos, videos, and games.
 - While using the Internet, keep personal data private and always verify the web links before opening them.
 - The World Wide Web (WWW) is a system of interconnected hypertext documents accessible via the Internet, with web pages forming websites that are accessible through web browsers like

Microsoft Internet Explorer, Google Chrome, Mozilla Firefox, Opera, and Safari.

- Web browsers allow various activities such as browsing websites, reading news, shopping, and social networking. To access a website, you simply enter the website's address in the browser's address bar.
- Search engines like Google help users find information quickly by typing keywords or phrases. For example, searching for "What is Artificial Intelligence" on Google returns relevant search results that users can filter or refine as needed.

Important Terms To Remember

Operating System: Software that helps control and manage computer's hardware/software and makes it easy for the user to use it.

Android: Android operating system is developed by Google Inc. It is the most widely used mobile phones' operating system.

iOS: iOS is developed by Apple Inc., for exclusive use on iPhones, iPads, and iPod Touch devices.

Taskbar: The taskbar is available at the bottom of the Windows desktop. It is an organization tool that organizes and provides quick access to various Windows features ranging from quick search or opening of applications to viewing and setting date and time, etc.

Icon: It is the graphical representation of any file or program.

Recycle Bin: The Recycle bin is a special type of system folder in Windows that temporarily stores all of your deleted files and folders. You can restore or recover all deleted items from the Recycle Bin.

File Explorer: You can browse and manage files and folders on your computer using the File Explorer. It allows you to create, copy, move, rename, and delete files and folders. It also allows you access to various drives and storage locations, like CD disk, USB, etc.

File: A file is an object that stores data and information. It is identified by its given name on a computer. The file can be any text file, document, picture, audio or video, etc

Folder: A folder is a container used to hold different types of files and sub-folders. Folders help in organizing and managing files.

PDF: PDF stands for Portable Document Format. This file format is supported by all operating systems.

Clipboard: It is a temporary memory. When you copy (or cut) anything, it is temporarily stored in the clipboard until pasted in desired location.

Network: A Network in computer science is a group of computers that are connected to each other through wired or wireless media to share their hardware and software resources and to exchange data and information among users.

Internet: The Internet is an international network consisting of millions of public, private and business networks. It connects people, places, and many other digital resources all around the world. Users can explore, learn and may easily connect with other users in order to share information or other interesting things, such as photos, videos, games, tutorials, etc.

WWW: WWW stands for World Wide Web. It is a data and information sharing system of interlinked hypertext documents, which are accessed via the internet.

Web Page: A single document shared on the Internet is called a web page.

Website: Collection or group of web pages is called a website. Websites contain information in form of documents, simple text, images, audio and video files.

Web Browser: A web browser is the software that provides you an interface to access the websites on the internet.

HTML: HTML stands for Hyper Text Markup Language. It is a markup language for creating web pages.

HTTP: HTTP stands for Hyper Text Transfer Protocol. It is the software used by the world wide web.

Search Engine: A search engine is a program that intelligently searches the information from other websites based on given keywords or phrases and returns you the list of websites links that contains these keywords or phrases.

Multi-user Operating System: A multi-user operating system like Unix or Linux is a type of operating system that allows multiple users to access a computer system concurrently.

Multi-tasking Operating System: A multi-tasking operating system, such as Windows, macOS, Unix and Linux, is a type of operating system that allows multiple tasks or programs to run concurrently on a computer.

Exercise

1. Encircle the correct answer

- i. The function of the Taskbar in Microsoft Windows 10 is to:
 - a. Manage computer resources
 - b. Provide user interface to create files and folders
 - c. Organize and provide quick access to various Windows features
 - d. Protect the computer from viruses
- ii. The function of the File Explorer in Windows 10 is to:
 - a. Manage computer resources
 - b. Organize and provide quick access to various Windows features
 - c. Browse and manage files and folders
 - d. Protects the computer from viruses
- iii. The location which temporarily stores the deleted files is called:

a. Windows Taskbar	b. Recycle Bin
c. System Tray	d. Windows Desktop

- iv. The function of the Quick Access Toolbar in MS Paint is to:
 - a. Draw images quickly
 - b. Insert shapes like circles, rectangles, etc
 - c. Draw images and artwork using freehand tool
 - d. Access commonly used features like Save and Undo, etc
- v. The purpose of a search engine on the Internet is to:
 - a. Help users to find information about specific topic
 - b. Help users to create web pages
 - c. Help users to download and watch movies
 - d. Help users to play online games

2. Fill in the blanks with appropriate words given below.

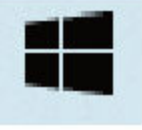
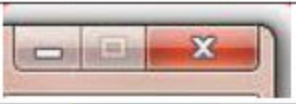
Keywords, CTRL+Z, Create shortcut, Notepad, Colours

- i. You can create a shortcut in Windows 10 by right-clicking on the selected file or folder, and then choose _____ in dropdown menu.
- ii. _____ is the Windows default text editor, which comes with all versions of Windows operating system.
- iii. The _____ palette shows a grid of colours for choosing the drawing lines and fill colours.
- iv. In MS Paint, you can undo the last action by pressing the _____ shortcut key.
- v. On the Internet, search engine helps its users to find required information or data based on given _____ or phrases.

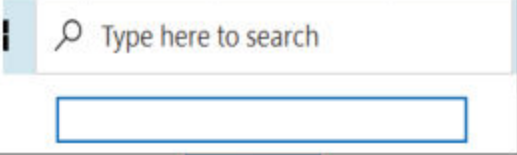
3. Provide the descriptive answers of the following questions.

- i. What is an Operating System?
- ii. What is the function of Taskbar in Windows 10?
- iii. What steps do you follow to outline or fill colours in drawings in Microsoft Paint?
- iv. List and describe some of the safety precautions while using the Internet?
- v. Explain the function of Search Engine.

4. Match the icons in Column A with Name in Column B.

Column A	Column B
	Windows 10 Start Menu / Start Button
	Recycle Bin
	Window Controls
	A Web Browser
	Brushes

5. Write the name of icon in the box provided and briefly describe in the next column.

Icon	Description
	
	
	



Note for Teachers

- Involve the students to explore Windows 10 operating system including the Microsoft (MS) Paint program and their main components by engaging them into hands on practice of Windows 10 operating system and MS Paint.
- Engage the students in hands on practice of searching notes and other material from different websites and online libraries.
- Engage the students in hands on practice of Search Engines and assign them the tasks like search for specific given topic using the different types of Search Engines, such as Google, Bing, Ask, etc, and then compare the returned results of each of these Search Engines.
- Engage the students in Task or Project Based Learning by assigning them tasks or projects in groups in order to enhance their team work skills. Assign tasks or projects related to Windows 10 operating system, MS Paint, and web surfing and search engines.
- Engage them to solve the textbook exercises to enhance their work skills.
- Use the hands-on practice strategy to cover the contents of this unit.